

Ludwig-Maximilian-Universität München - Faculty of Physics

Photonics and Optoelectronics - THz Group

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Methoden zur Schichtdickenbestimmung von Parylene Dünnschichten

während chemischer Gasphasenabscheidung

Bachelor Arbeit

von Matthieu Bernard Michel Mézière

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Ludwig-Maximilian-Universität München - Faculty of Physics

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Parylene Thin-Film Thickness Monitoring Methods
during Chemical Vapour Deposition

Bachelor Thesis

by Matthieu Bernard Michel Mézière

submitted on 14.01.2024

Abstract

Construction of a quartz crystal microbalance using ordinary components obtainable from electronic suppliers has been performed. Off the shelf quartz crystals which are generally used in consumer electronic devices have been chosen as sensors. The microbalance built is fully functional, however, further work is needed to improve its accuracy. For the time being, use of the previously known optical monitoring method relying on optical thin film interference is preferable.

Acknowledgements

This work has been conducted at the THz Group, part of the chair of photonics and optoelectronics of the Ludwig-Maximilian university in Munich, under the direction of Prof. Dr. Roland Kersting.

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I wish also to thank Johannes Steinberger, Tianyi Cao and Md. Hanif, my colleagues at the THz group, for their help when operating the CVD machine.

There never was a dull moment at the THz group, and it has been a pleasure to be part of it.

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1 Introduction

The THz Group conducts research on organic semiconductors for which metal-insulator-semiconductor (MIS) capacitors are fabricated. The insulator consists of a 350 nm thick layer of a polymer known as parylene N. It is applied to the substrate via chemical vapour deposition (CVD). The thickness of the insulator is monitored using an optical method. Light reflected in the samples changes colour while the film grows. The film thickness can be deduced from the observed colours.

Human colour perception is somewhat subjective, and the question arises whether the film thickness could be determined more precisely.

Devices known as quartz crystal microbalances are often used for film thickness monitoring, and are very precise. They rely on a quartz crystal to sense the amount of material deposited. Commercial quartz crystal microbalances are somewhat expensive, and use crystals specifically fabricated for the purpose.

In the course of this bachelor thesis, a quartz crystal microbalance in its most basic form is built using parts widely available from electronics suppliers such as quartz crystals intended for use in watches or radios.

Before addressing the “do it yourself” microbalance, chemical vapour deposition of parylene is described in detail, as well as the underlying optical principles of the thickness monitoring method used until now at the THz Group. The basic operation principles of quartz crystal microbalances are then presented, followed by design guidelines for oscillators driving quartz crystals.

Having given an overview of the underlying theory, the THz Group’s CVD machine is then described in detail. Determination of film thickness using a spectrophotometer is then presented, with which reference measurements are obtained. In the course of the experiments conducted, a device to mask the substrate being coated during CVD was required to gain insight on deposition rates. The layout of this device is described. Finally, the “do it yourself” quartz microbalance featured in the experimental section is detailed.

The behaviour of the crystals during deposition of parylene is studied, in order to determine whether such crystals are suitable as microbalance sensors. In parallel, the optical monitoring method is revised in order to confirm its accuracy.

2 Theoretical Background

2.1 Chemical vapour deposition (CVD) of parylene-N

Chemical vapour deposition is a method to produce organic thin films. The substrate is placed into an atmosphere of gaseous precursors, which gradually polymerize on the substrate's surface.

A polymer that is commonly produced via chemical vapour deposition is poly-para-xylylene, also known as parylene (figure 1).

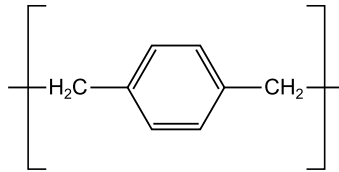


Figure 1: Repeating unit of parylene N.

Parylene is chemically inert, stable up to high temperatures and has a high dielectric strength. Parylene films are pin-hole free and conformal to the substrate, without thinning at the edges [2, Sec. 3.1.2]. The polymerization route via chemical vapour deposition is known as the Gorham process. No byproducts are created during polymerization, and only one precursor is required [1, Sec 5.4].

The precursor is di-para-xylylene, the parylene dimer [1, Sec. 4.1]. The Gorham process takes place in medium vacuum ($p \lesssim 40 \mu\text{bar}$).

First, the parylene dimer is sublimated at a temperature of $T = 120^\circ\text{C}$. The gaseous dimer is then cracked into monomers (para-xylylene) at a temperature of $T = 660^\circ\text{C}$ (figure 2) [1, Sec. 1.4.1]. The monomer may exist in two different structures, the quinoid and benzoid forms. The quinoid form is energetically preferred, only a small fraction of the monomers have the diradical structure [1, Sec. 4.2].

Parylene chain initiation (figure 3) takes place at room temperature once monomers have adsorbed to the surface of the substrate [1, Sec 1.4.1]. Polymerization rate increases with decreasing temperature [1, Sec. 5.4.2]. Three gaseous monomers react to form a solid diradical oligomer. The chain grows one monomer at a time [1, Sec. 5.4].

After deposition, the parylene layer consists of diradical chains. Once the coated substrate is returned to normal atmosphere, oxygen diffuses into the parylene film. The oxygen reacts with the radical chain ends, terminating the parylene chains [1, Sec. 1.2].

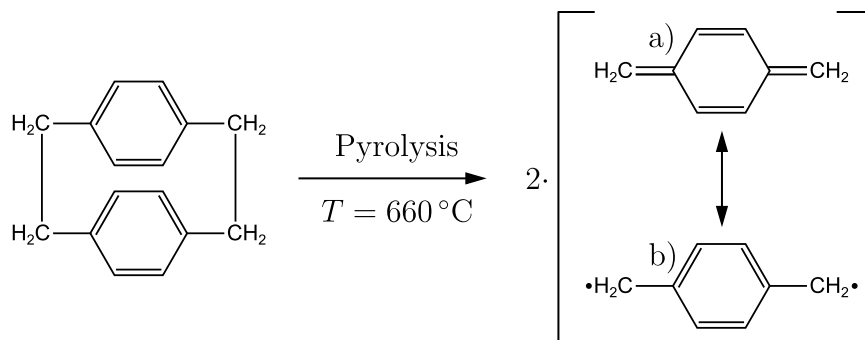


Figure 2: The sublimated dimers are cracked into monomers. Two different monomer structures are possible: a) Quinoid form, preferred - b) Diradical or benzoid form.

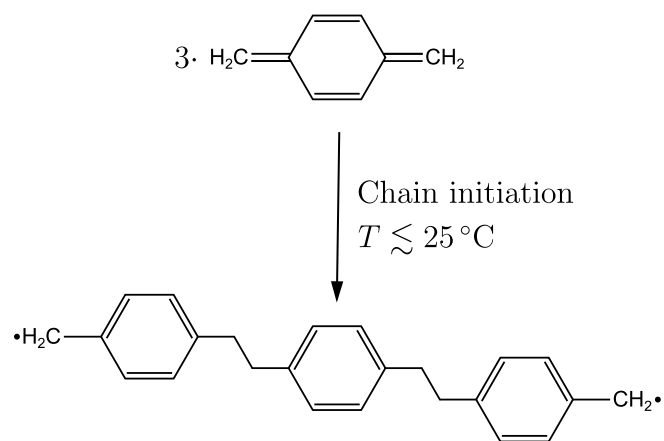


Figure 3: Parylene chain initiation. Three monomers react to form a diradical oligomer.

2.2 Monitoring film thickness - optical method

The thickness of the growing film can be monitored during CVD by observing light reflected by the sample. Due to interference phenomena, only certain wavelengths of light are reflected at a certain film thickness. The film thickness can then be deduced from the observed colours.

2.2.1 Reflexion at interfaces

When light of intensity I_0 encounters the the interface between two different optical media, part of the light is reflected (I_r), while the remainder is refracted (I_t):

$$I_0 = I_r + I_t \quad (1)$$

This continuity equation can be rewritten using the intensity ratios, introducing the transmittance T and reflectance R .

$$\frac{I_t}{I_0} + \frac{I_r}{I_0} = T + R = 1 \quad (2)$$

Both reflectance and transmittance can be calculated using the Fresnel equations. The Fresnel equations take the inbound light's polarization into account. The polarization is denoted with respect to the plane of incidence, the plane in which the incident and reflected beams lie. The electric field of p-polarized light lies in the plane of incidence, for s-polarized light, it lies in a plane perpendicular to the plane of incidence.

$$R_s = |r_s|^2 = \left(\frac{n_1 \cos(\alpha) - n_2 \cos(\beta)}{n_1 \cos(\alpha) + n_2 \cos(\beta)} \right)^2 \quad (3)$$

$$R_p = |r_p|^2 = \left(\frac{n_2 \cos(\alpha) - n_1 \cos(\beta)}{n_2 \cos(\alpha) + n_1 \cos(\beta)} \right)^2 \quad (4)$$

For unpolarized light, the effective reflectance R is the average of R_s and R_p :

$$R = \frac{1}{2} \cdot (R_s + R_p) \quad (5)$$

The reflectances depend on the incident angle α and the refractive indices n of both

materials. The refraction angle β is given by Snell's law:

$$n_1 \sin(\alpha) = n_2 \sin(\beta) \quad (6)$$

Refractive indices are complex valued and consist of the real part n and the imaginary part κ , also known as the extinction coefficient. For transparent dielectrics, $\kappa \ll n$, so their refractive index can be treated as real valued. However, this is not the case for metals ($n \approx \kappa$), which explains their high reflectance.

2.2.2 Absorbtion

Light travelling a distance d through a medium loses intensity due to scattering and absorbtion. The Lambert-Beer law describes this phenomenon.

$$I_t(d) = I_t(0) \cdot \exp(-\mu(\lambda) d) \quad (7)$$

The absorbtion coefficient $\mu(\lambda)$ is related to the extinction coefficient κ :

$$\mu(\lambda) = \frac{4\pi \kappa}{\lambda} \quad (8)$$

The (complex) refractive index may also be wavelength dependent. Materials that are transparent in the visible spectrum may therefore absorb certain wavelengths. Glass, for instance, absorbs UV-radiation.

2.2.3 Phase shifts

At the interface to an optically denser material ($n_2 > n_1$), “hard” reflection occurs: the phase of the reflected light shifts by $\phi = \pi = \frac{\lambda}{2}$.

Light reflected off an absorbing material ($\kappa \neq 0$) may recieve a different phase shift ($0 \leq \phi \leq \pi$). It can be calculated from the Fresnel coefficients r_p and r_s :

$$\phi = \arctan\left(\frac{\text{Im}(r_p)}{\text{Re}(r_p)}\right) \quad (9)$$

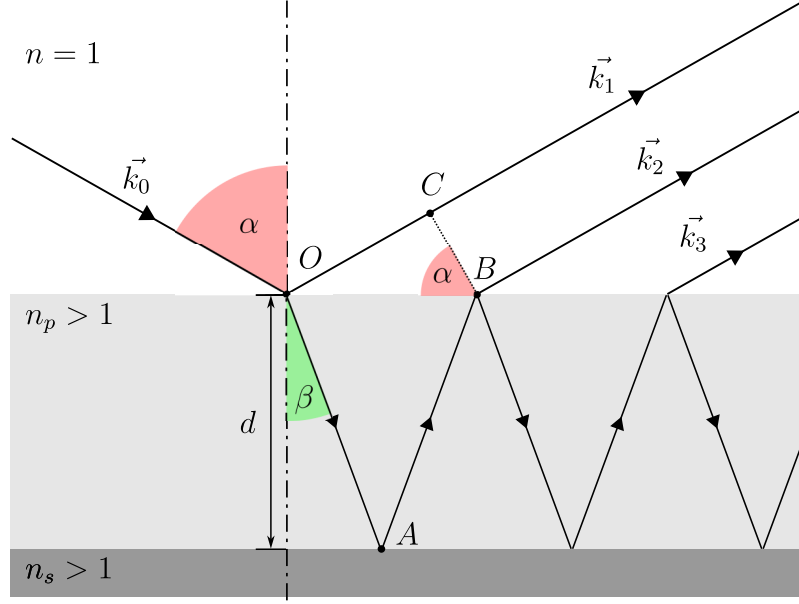


Figure 4: Multiple reflections take place at the parylene film boundaries. The reflected beams $\vec{k}_j$ interfere with each other.

2.2.4 Multiple reflections

Figure 4 shows the situation when monitoring the parylene film thickness during CVD. The parylene is deposited on a chromium substrate with refractive index n_s . A white light source is directed at the substrate. Multiple reflections take place inside the parylene film. The intensities I_j of the outbound beams $\vec{k}_j$ are given by:

$$\frac{I_1}{I_0} = R_{pv} \quad (10)$$

$$\frac{I_{j>1}}{I_0} = T_{vp} \cdot R_{ps}^{j-1} \cdot R_{pv}^{j-2} \cdot T_{pv} = (1 - R_{pv})^2 \cdot R_{pv}^{j-2} \cdot T_{pv} \quad (11)$$

Figure 5 shows the reflectance $R_{pv} = R_{vp}$ of the parylene-vacuum interface for all possible incidence angles α , using $n_p = 1.66$ [1, tab. 6.1]. In practice, $0^\circ < \alpha < 45^\circ$, so the reflectance is $R_{pv} \approx 0.06$. Since the substrate is metal, the reflectance of the parylene-substrate interface can be estimated to be $R_{ps} \approx 0.9$. The intensity ratios are then:

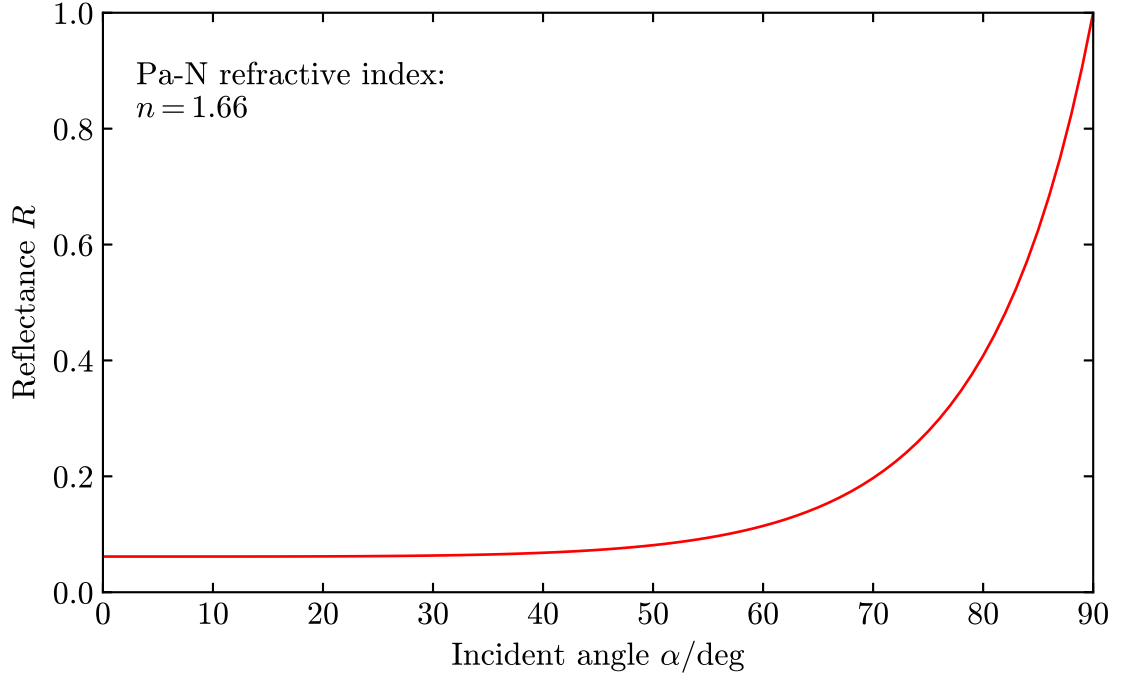


Figure 5: Reflectance of the parylene-vacuum interface. In practice, the incident angle $\alpha \approx 30^\circ$, yielding a reflectance of $R_{pv} \approx 0.06$.

$$\frac{I_1}{I_0} \approx 0.06 \quad (12)$$

$$\frac{I_2}{I_0} \approx 0.94^2 \cdot 0.9 = 0.80 \quad (13)$$

$$\frac{I_3}{I_0} \approx 0.80 \cdot 0.06 \cdot 0.9 = 0.04 \quad (14)$$

$$\frac{I_4}{I_0} \approx 0.04 \cdot 0.06 \cdot 0.9 = 0.002 \quad (15)$$

$$\frac{I_{j>4}}{I_0} \longrightarrow 0 \quad (16)$$

2.2.5 Interference

Intensities after the third internal reflection ($j > 3$) become very small. Therefore, major (visible) interference phenomena will only arise from interference of $\vec{k}_1$, $\vec{k}_2$ and $\vec{k}_3$.

Constructive interference occurs only when the following condition is fulfilled:

$$\Delta L = L_2 - L_1 = m \lambda \quad m \in \mathbb{N} \quad (17)$$

The optical path length difference ΔL must be a whole numbered multiple of the wavelength λ . The individual optical path lengths L_j are deduced from geometrical considerations. Phase shifts ϕ due to reflection add to the optical path length.

$$L_1 = \overline{OC} + \phi_{vp} = \sin(\alpha) \cdot \overline{OB} + \phi_{vp} = 2d \cdot \sin(\alpha) \cdot \tan(\beta) + \phi_{vp} = \frac{2d n_p}{\cos(\beta)} \cdot \sin^2(\beta) + \frac{\lambda}{2} \quad (18)$$

$$L_2 = n_p \cdot \overline{OAB} + \phi_{ps} = \frac{2d n_p}{\cos(\beta)} + \phi_{ps} \quad (19)$$

$$L_3 = 2 \cdot L_2 + \phi_{pv} \quad (20)$$

The following conditions for constructive interference can be derived:

$$\Delta L_{12} = L_2 - L_1 = \frac{2d n_p}{\cos(\beta)} \cdot (1 - \sin^2(\beta)) + \phi_{ps} - \phi_{vp} = m \lambda \quad (21)$$

$$\Delta L_{13} = L_3 - L_1 = \frac{2d n_p}{\cos(\beta)} \cdot (1 + \cos^2(\beta)) + \phi_{pv} - \phi_{vp} = m \lambda \quad (22)$$

$$\Delta L_{23} = L_3 - L_2 = L_2 + \phi_{pv} = \frac{2d n_p}{\cos(\beta)} + \phi_{pv} - \phi_{ps} = m \lambda \quad (23)$$

2.2.6 Determining film thickness

Assuming a $\phi_{ps} = \frac{\lambda}{2}$ phase shift at the parylene-chromium interface, three possible formulas for the film thickness are obtained:

$$d_{12} = \frac{m}{2 n_p \cos(\beta)} \cdot \lambda \quad (24)$$

$$d_{13} = \frac{2m+1}{4 n_p} \cdot \frac{\cos(\beta)}{1 + \cos^2(\beta)} \cdot \lambda \quad (25)$$

$$d_{23} = \frac{2m+1}{4 n_p} \cdot \cos(\beta) \cdot \lambda \quad (26)$$

Judging by the intensities of the interfering beams, equation 24 is most likely to apply to visible interference phenomena.

2.3 Quartz crystal microbalances (QCM)

Quartz crystal microbalances use a piezoelectric resonator to sense the film thickness. When material is deposited on the resonator, its mechanical properties are altered. This causes the resonance frequency to change. The mass of the added material can then be deduced from the frequency shift.

2.3.1 Piezoelectric effect

In some crystalline dielectrics, the dipoles present in the unit cells are all orientated in one direction, resulting in a natural, permanent polarization. Applying mechanical stress to the crystal displaces the dipoles, altering the polarization. Polarization charges appear at the surfaces of the crystal, creating a potential difference between the surfaces [4, Sec. 11-6]. The effect is reversible: A piezoelectric crystal will deform when subjected to an electric field.

2.3.2 Piezoelectric resonators

Two opposite faces of a piezoelectric crystal are plated with metal electrodes. The crystal can be driven into oscillation by applying an alternating voltage $u_d(t)$ between the electrodes. The induced displacement $x(t)$ will in turn cause an alternating voltage $u(t)$ to appear between the electrodes. When mechanical resonance occurs, the voltage generated by the crystal $u(t)$ is then in phase with $u_d(t)$, electrical resonance occurs.

The situation can be modelled using the electrical equivalent of a damped mechanical oscillator, a series LCR circuit (figure 6). The electrodes and the crystal form a capacitor, which is represented by the parallel capacitance C_e [5, p. 444].

The impedance of the network $Z(\omega = 2\pi f)$ is given by:

$$Z(\omega) = \frac{Z_m \cdot Z_e}{Z_m + Z_e} = \frac{R_m - \frac{i}{\omega C_e} \left(\omega L_m - \frac{1}{\omega C_m} \right)}{R_m + i \left(\omega L_m - \frac{1}{\omega C_m} \right) - \frac{i}{\omega C_e}} \quad (27)$$

Figure 7 shows the frequency response of the network, calculated using SPICE simulation.

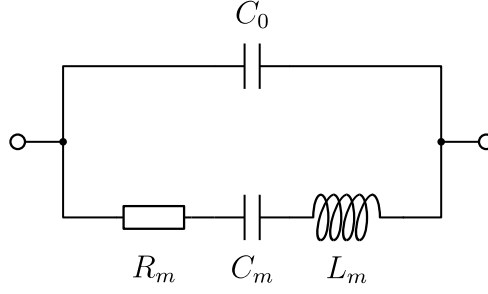


Figure 6: Butterworth-Van Dyke electrical equivalent circuit of an ideal quartz resonator. L_m models the motional mass, C_m the material's elasticity, R_m mechanical losses. C_0 describes the capacitance between the electrodes. [3, p. 164]

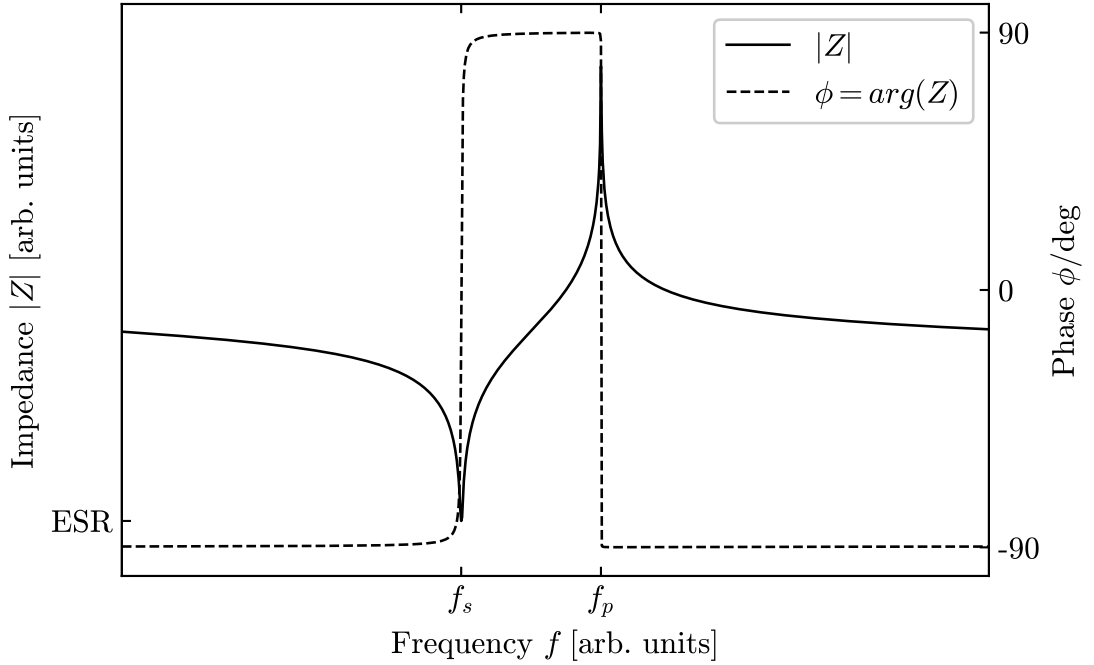


Figure 7: Frequency response of the Butterworth-Van Dyke network.

Two sharp resonances are present at f_s and f_p . The network is capacitive (phase lags by $\phi = -90^\circ$) until f_s is reached. At f_s , the LCR branch is in resonance and is purely resistive ($\phi = 0$) [5, p. 444]. The impedance minimum at series resonance f_s (equation 28, [9, eq. 5]) is equal to the equivalent series resistance (ESR) specified in the crystal's datasheet.

$$f_s = \left(2\pi \sqrt{L_m \cdot C_m}\right)^{-1} \quad (28)$$

The LCR branch becomes inductive for $f_s < f < f_p$, phase leads by $\phi = 90^\circ$. This is

known as the parallel resonance region. At f_p (equation 29, [9, eq. 6]), the inductive LCR branch and the capacitance C_e form a parallel LC resonant circuit, and impedance peaks [5, p. 444].

$$f_p = f_s \cdot \left(1 + \frac{C_m}{2C_e}\right) \quad (29)$$

Quartz crystals are commonly used as piezoelectric resonators. Cutting the resonator from the raw crystal along a certain crystallographic axis offers good temperature stability at room temperature. This particular cutting direction is known as the AT-cut. AT-cut quartz crystals perform a shearing oscillation perpendicular to the electric field as shown in figure 8 [6, p. 3].

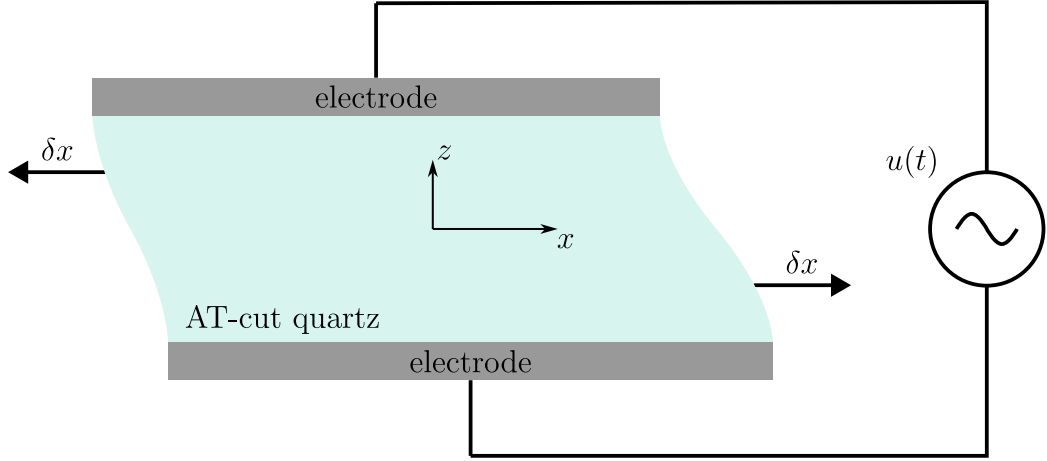


Figure 8: AT-cut quartz performs a shearing motion perpendicular to the applied electric field. At resonance, the induced acoustic wave becomes stationary, with $\lambda = 2d_q$.

The resonant frequency depends on the geometry of the resonator. Resonators consisting of an AT-cut quartz plate as shown in figure 8 are available in frequencies in the MHz range. Since the crystal is sheared across its thickness, mechanical resonance occurs when the induced acoustic wave's wavelength λ equals twice the crystal's thickness d_q [6, p.4].

$$d_q = \frac{\lambda}{2} \quad (30)$$

The resonant frequency is then given by equation 31, where v is the acoustic wave propagation speed in the crystal.

$$f_0 = \frac{v}{\lambda} = \frac{v}{2d_q} \quad (31)$$

Such resonators are not completely flat, but are given a convex shape in order to confine the oscillation to the center of the crystal [7]. This allows the crystal to be mounted by the edges without disturbing the resonator.

To achieve lower frequencies, the quartz crystal is cut into the shape of a tuning fork. These quartz tuning forks are commonly used in watches and are tuned to a resonant frequency of $f_0 = 32.768 \text{ kHz} = 2^{15} \text{ kHz}$. The oscillation of a tuning fork can be modelled by considering each prong as a flexing beam (Euler-Bernoulli beam). The fundamental oscillation mode is shown in figure 9. The prongs flex towards and away from each other [8].

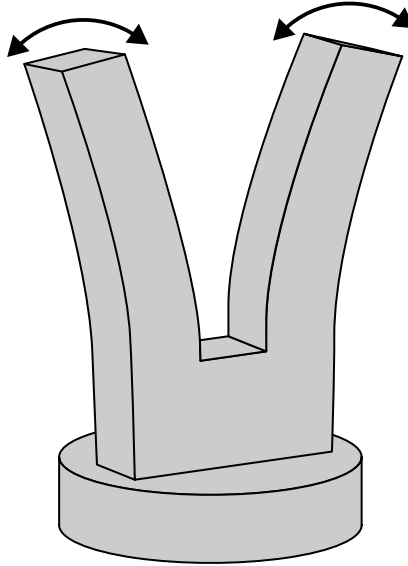


Figure 9: Fundamental oscillation mode of tuning fork. Flexure is highly exaggerated.

2.3.3 Depositing material - Sauerbrey equation

The resonance frequency of a quartz crystal resonator is given by equation 31 and depends on the thickness of the crystal. Differentiating equation 31 by d_q yields the frequency shift caused by a change in quartz thickness Δd_q .

$$\Delta f = \frac{\partial f}{\partial d} \Delta d_q = -\frac{v_q}{2d_q^2} \Delta d_q \quad (32)$$

A change in the crystal thickness d_q is equivalent to loading the crystal with an additional mass Δm , A_q being the surface of the crystal and $\rho_q = 2.65 \text{ g} \cdot \text{cm}^{-3}$ the density of quartz [9, p.209]:

$$\Delta m = A_q \rho_q \Delta d_q \quad (33)$$

Using the relation above to rewrite equation 32 and inserting $d_q = \frac{v_q}{2f_0}$ gives the Sauerbrey equation [9, eq. 4a]:

$$\Delta f = -\frac{v_q}{2 \left(\frac{v_q}{2f_0} \right)^2} \cdot \Delta d_q = -\frac{2f_0^2}{v_q} \cdot \Delta d_q = -\frac{2f_0^2}{v_q} \cdot \frac{\Delta m}{A_q \rho_q} \quad (34)$$

The mass Δm of material deposited on a quartz resonator with initial frequency f_0 can be deduced from the frequency shift Δf . The quartz crystal resonator functions as a weighing scale, hence the designation “quartz crystal microbalance”. The term $\frac{v_q}{2} \equiv N = f_0 \cdot d_q$ is also known as the frequency constant. For AT-cut quartz, $N = 1670 \text{ KHz mm}$ [9, p. 209].

To determine the thickness d_x of a material of density ρ_x , the Sauerbrey equation is rewritten using the relation

$$\Delta d_q = \frac{\rho_x}{\rho_q} d_x \quad (35)$$

which gives:

$$d_x = \frac{N}{f_0^2} \cdot \frac{\rho_q}{\rho_x} \cdot \Delta f \quad (36)$$

However, there is a limitation to the Sauerbrey equation. The material deposited on the quartz crystal may have a different acoustic impedance, which is neglected in Sauerbrey’s approach. The coated resonators’ behaviour is adequately modelled only for thin layers [6, p. 12]:

$$\frac{\Delta m}{m_q} = \frac{\rho_x d_x}{\rho_q d_q} \lesssim 2\% \quad (37)$$

For parylene-N ($\rho_p = 1.11$), this 2% limit corresponds to a film thickness of $d \approx 47 \mu\text{m}$:

$$d_p \lesssim 0.02 d_q \cdot \frac{\rho_q}{\rho_p} = 0.02 \text{ mm} \cdot \frac{2.65}{1.11} = 47.75 \mu\text{m} \quad (38)$$

At the THz group, film thicknesses of $d_p \approx 350 \text{ nm}$ are desired for the MIS-capacitor insulator layer. The Sauerbrey equation is therefore perfectly suitable for film thickness determination.

2.4 Oscillators

An (electrical) oscillator essentially consists of an amplifier, to which positive feedback is applied, but only at the desired oscillation frequency. Quartz crystal resonators are excellent frequency selective elements. Placed in the feedback loop of an inverting amplifier, the crystal can provide the required positive feedback required to drive the amplifier into oscillation.

The Pierce topology follows this approach, and can be implemented using a CMOS inverter as an amplifier. Figure 10 shows the schematic of such an oscillator. The CMOS inverter is biased in its linear region by R_B .

A phase shift of $\phi = 360^\circ$ between in- and output of the amplifier is required for positive feedback. Using an inverting amplifier, the feedback loop must provide 180° phase shift. However, in its parallel resonance region, phase in the crystal X leads by only 90° . Adding C_1 and C_2 forms a π -network with the crystal, adding another 90° to the crystal's phase shift. The condition for oscillation is now fulfilled.

The series capacitance C_L of the two capacitors C_1 and C_2 is given by:

$$C_L = \frac{C_1 \cdot C_2}{C_1 + C_2} \quad (39)$$

C_L is effectively in parallel with the crystal, increasing the capacitance C_e in the Butterworth Van Dyke model (figure 6). Therefore, the oscillators frequency can be tuned by altering C_1 and C_2 , which is known as “frequency pulling”. C_L is the load capacitance of the crystal. Crystals oscillate at their specified frequency only for a certain load capacitance C_L specified in the datasheet.

The resistor R_S limits the crystal's drive current. R_S is usually selected equal to C_2 's impedance:

$$R_S = Z_{C_2} = (\omega_X \cdot C_2)^{-1} = (2\pi f_0 \cdot C_2)^{-1} \quad (40)$$

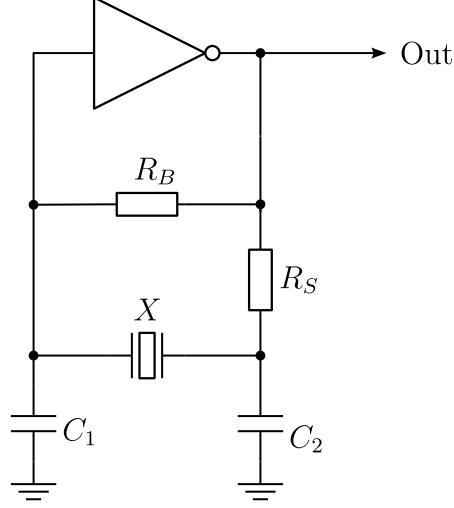


Figure 10: Pierce oscillator using a CMOS inverter as an amplifier. R_B biases the inverter in its linear region. The crystal X provides positive feedback at its resonance frequency f_0 . The capacitors C_1 and C_2 load the crystal. R_S limits the current driving the crystal and forms a low pass filter with C_2 to prevent oscillation at undesired modes.

3 Experimental Setups

3.1 Chemical vapour deposition machine

3.1.1 Overview

The basic layout of the CVD machine is shown in figure 11.

Operating pressures of $p = 10 \dots 30 \mu\text{bar}$ are achieved using a rotary vane pump (10). All pipes between the sublimation furnace (2) and the deposition chamber (5) are heated to a temperature of $T_p = 120^\circ\text{C}$. A cold trap (9) filled with liquid nitrogen prevents parylene from reaching the pump. The substrate holder (6) can be heated up to $T_s = 90^\circ\text{C}$ with halogen lamps to eliminate water contaminations of the sample. The substrate holder is also fitted with a water cooling system. This allows the substrate to be cooled to $T_s = 11^\circ\text{C}$, favouring parylene polymerization.

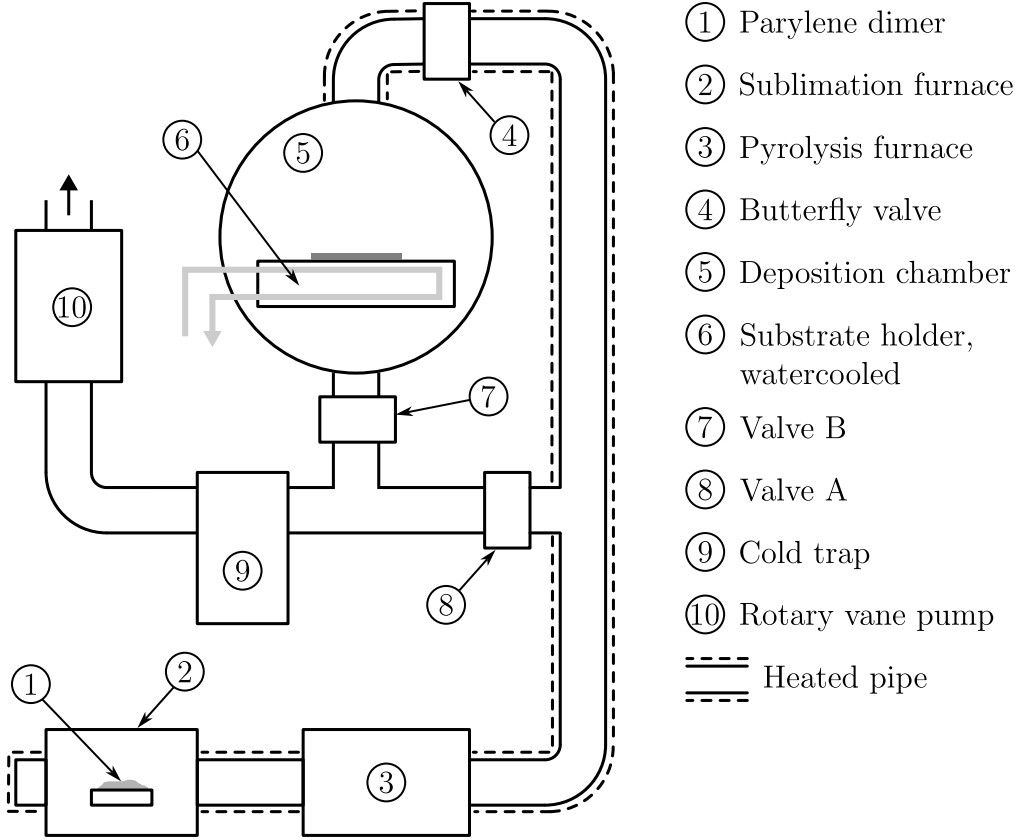


Figure 11: Layout of the THz Group's CVD machine.

3.1.2 Deposition process

Parylene dimer ($m_p \approx 1.5$ g) is loaded in the sublimation furnace (2). The system is evacuated. The butterfly valve (4) is shut, closing off the deposition chamber. Valves A and B (7,8) are open. The pipes are heated. The substrate holder is also heated. When its temperature reaches $T_s = 90^\circ\text{C}$, the watercooling is turned on. Once the pipes reach their operating temperature ($T_p = 120^\circ\text{C}$), the pyrolysis furnace (3) is heated to $T_2 = 660^\circ\text{C}$. This prevents any parylene dimer from reaching the deposition chamber. The sublimation furnace (2) is heated last ($T_1 = 120^\circ\text{C}$). Parylene dimer begins to evaporate and travels to the pyrolysis furnace. The dimer is split into monomers. While the butterfly valve is shut, the monomers are discarded in the cold trap (9). Deposition is initiated 30 min after the sublimation furnace reaches its operating temperature. The butterfly valve (4) is opened and valve A (8) is closed. The monomers now travel to the deposition chamber. At this point, the substrate has been cooled to a temperature below room temperature. The monomers polymerize on the substrate's surface. Monomers escaping the deposition chamber are caught in the cold trap. When the desired film thickness has been reached,

the butterfly valve is closed and valve A (8) is reopened. Parylene deposition ceases immediately. The sublimation furnace is shut down first. At a temperature of $T_1 \leq 90^\circ\text{C}$, the parylene dimer ceases to evaporate. The pyrolysis furnace is turned off. When its temperature is below $T_2 = 400^\circ\text{C}$, the pipe heating is turned off.

The machine is flooded only the following day, when the cold trap has returned to room temperature. This prevents water contained in the air from condensating inside the cold trap. Condensed water would diffuse in the parylene layer lining the trap, and adsorb to the trap's inner surfaces. During the following evacuation of the machine, water would be slowly released into the vacuum, spoiling it.

Since the machine is evacuated during the deposition process, heat is only transferred via radiation. The interior of the deposition chamber is at room temperature. No heat is transferred by the parylene monomers. This was confirmed by placing a PT1000 sensor inside the chamber during a deposition.

3.2 Spectrophotometer

Estimating the film thickness by observing the reflection colours on the sample is not accurate, since human colour perception is rather subjective. This is the method used until now to determine the film thickness, using equation 24 with $\beta = 0^\circ$.

However, the achieved film thickness can be determined more accurately after deposition by measuring the reflectance spectrum of the coated chromium reflector.

The spectrum is measured using a Shimadzu UV-1800 spectrophotometer. Figure 12 shows the layout of the machine. The intensity of light transmitted through both samples is measured for wavelengths ranging from 1100 nm – 190 nm.

Figure 13 shows the reflectance spectrum of a parylene coated reflector.

The peaks show at which wavelengths λ_{max} constructive interference occurs. The valleys correspond to destructive interference. The smaller the extremum's wavelength, the higher the interference order. The film thickness can be determined using the wavelengths of both the maxima (equation 24, with $\beta = 0^\circ$) and minima (λ_{min}). For the minima, the equation is adapted:

$$d = \frac{2m + 1}{4n_p} \cdot \lambda \quad (41)$$

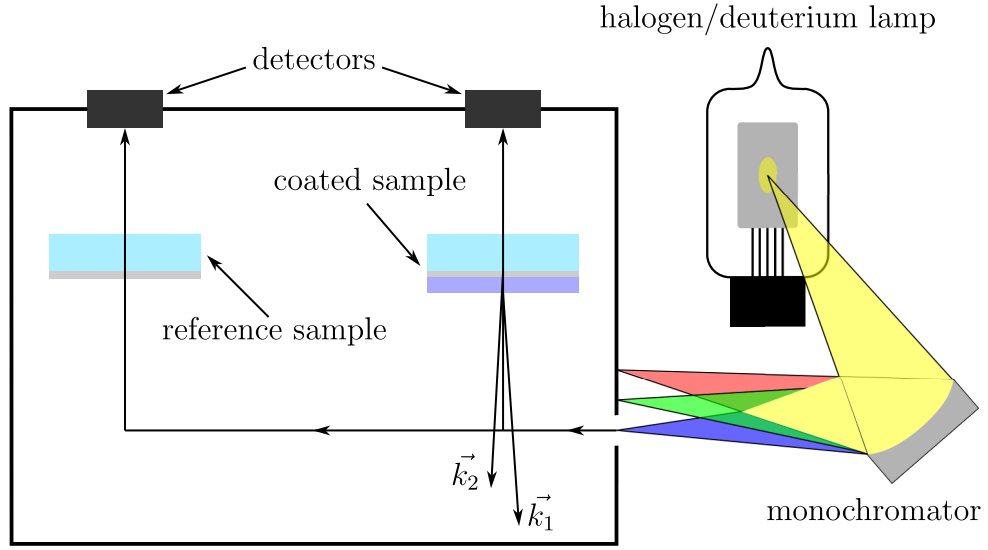


Figure 12: Basic layout of the Shimadzu UV-1800 Spectrophotometer. The device performs differential measurements versus a reference sample. Here, reflectance of a chrome reflector coated with parylene is measured, in order to determine parylene layer thickness. Reflected beams $\vec{k}_1$ and $\vec{k}_2$ correspond to those shown in figure 4.

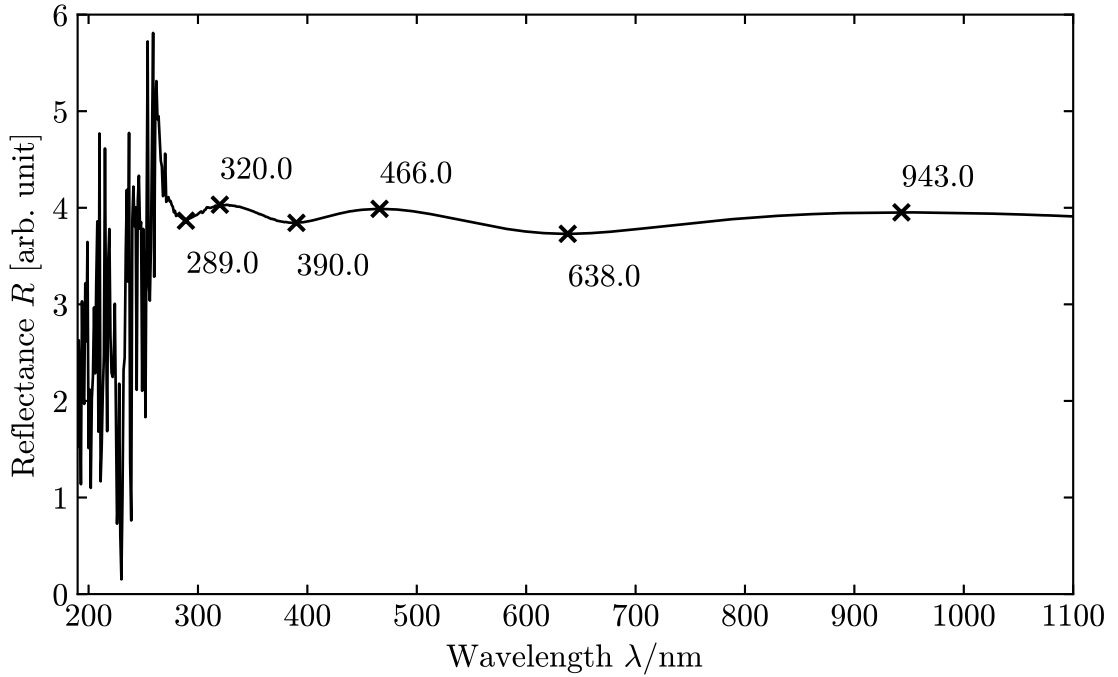


Figure 13: Reflectance spectrum of a parylene coated reflector, measured as shown in figure 12. Layer thickness can be determined from extrema wavelengths.

The coated sample need not be highly reflective. Usable spectra are also obtained from parylene coated microscope slides. However, different equations must be used. The mi-

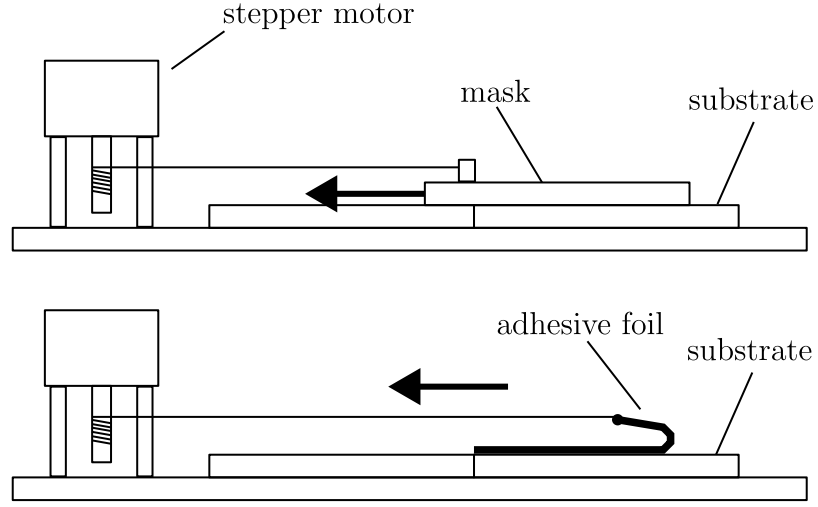


Figure 14: Masking device used to expose different areas of a microscope slide to the parylene atmosphere for different times. A stepper motor forms a winch, which pulls on a mask. The adhesive foil version works best.

microscope slides are made of soda-lime glass ($n_s \approx 1.5$). Since $n_s < n_p$, “soft” reflection occurs at the parylene-substrate interface. Therefore, no phase change occurs for reflections inside the parylene layer ($\phi_{ps} = 0$). The film thickness for parylene coated glass is then given by:

$$d = \frac{2m+1}{4n_p} \cdot \lambda_{max} \quad (42)$$

$$d = \frac{m}{2n_p} \cdot \lambda_{min} \quad (43)$$

At wavelengths below 270 nm, the reflectance spectrum XXX

3.3 Substrate masking device

In order to investigate deposition rate during CVD, different areas of the substrate could be exposed or masked off at different times. The substrate would have a parylene layer of different thicknesses. The deposition rate could then be determined from the thickness and exposure time of each area. A device is conceived to accomplish this.

The masking device is shown in figure 14. A wire is wound around the axle of a stepper motor to make a winch. The end of the wire is fastened to a microscope slide. This

microscope slide acts as a mask. The mask lies on top of another microscope slide, the sample slide. When the motor turns, it pulls the mask across the sample slide, uncovering it.

Unfortunately, this approach was unsuccessful. Parylene seeps under the mask.

The mask is replaced by adhesive foil, which is stuck onto the sample slide. The winch pulls onto the foil, peeling it off the sample. This gives excellent results.

3.4 The “poor scientist’s” quartz crystal microbalance

A simplistic quartz crystal microbalance is set up using commercial quartz crystals sold for electronic applications. The crystals are driven by a Pierce oscillator. Oscillator frequency is measured using a Keithley 2000 multimeter, which has a frequency counter function.

3.4.1 Quartz crystals

32.768 kHz tuning forks (Abracon AB38T) and 1.8432 MHz convex crystals (ECS-018-S-1X and IQD LFX TAL003033) are purchased. The crystals come in a sealed casing, which must be opened to expose the resonator.

The tuning forks are covered with a cylindrical cap, which can be easily removed using pliers. This is done by squeezing the cap at the crystal’s base several times. After each squeeze, the crystal is rotated. At some point, either the base will crumble away, or the cap will have widened enough to slide off, revealing the tuning fork. The tuning fork is usable in both cases.

The convex crystals’ caps are welded to the base. The only way to open these is to cut the cap off at the base using a hacksaw or a cutting disc.

3.4.2 Oscillator circuits

Two Pierce oscillator circuits (figure 15) are built to drive the 32.768 kHz tuning forks and the 1.8432 MHz convex crystals.

The CD4060 IC is sold as a “14-stage ripple carry binary counter”. It contains an inverter specifically designed to be used as an oscillator. The counter portion of the IC can be used to perform frequency division of the signal. Table 1 shows the component values for both versions of the circuit.

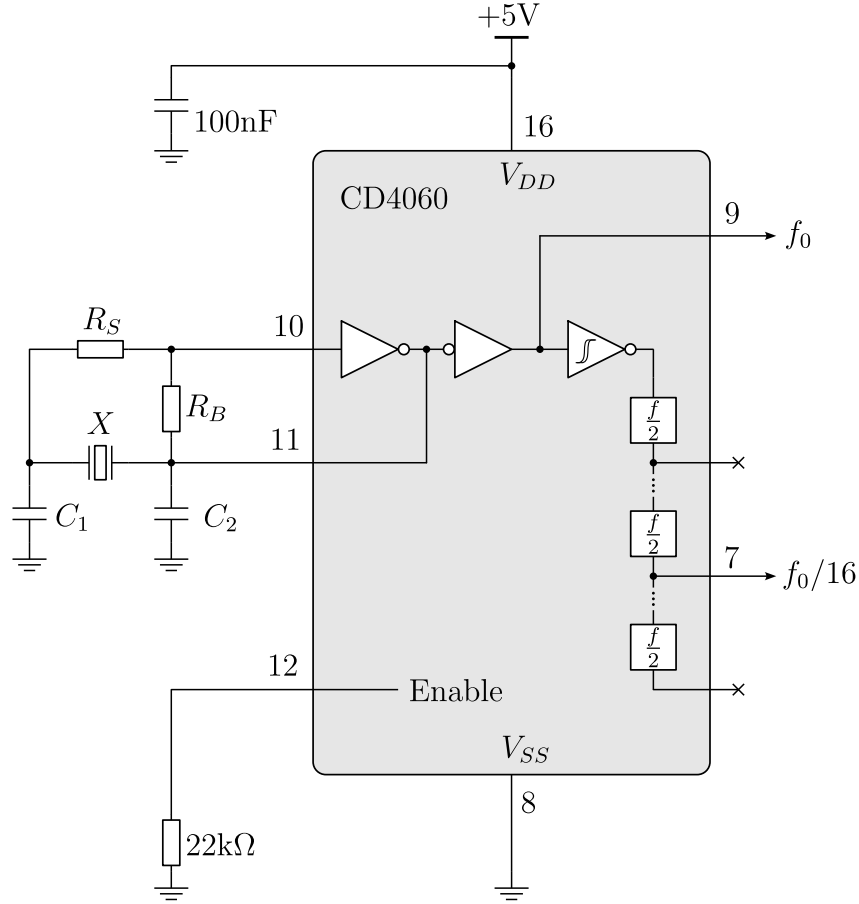


Figure 15: Schematic of the Pierce oscillator used to drive the 32kHz tuning forks and 1.8MHz convex crystals. The CD4060 IC is designed to function as a pierce oscillator. A flip-flop array allows frequency division of output signal.

f_0	1.8432 MHz	32.768 kHz
R_B	1 M Ω	10 M Ω
R_S	5.1 k Ω	470 k Ω
C_1	20 pF	25 pF
C_2	20 pF	25 pF

Table 1: Component values for the Pierce oscillator circuit shown in figure 15.

The output signals of the 1.8432 MHz oscillators are divided by 16 to suit the frequency counter at hand (Keithley 2000 multimeter), which can handle frequencies up to 350 kHz only.

The crystals should not be placed too far from the oscillator. For the 1.8432 MHz crystals, 20 cm wires leading to the crystal pose no problems.

Up to eight different crystals can be driven with one oscillator using two SN74LV4051A multiplexer ICs to switch the crystals to the oscillator (figure 16).

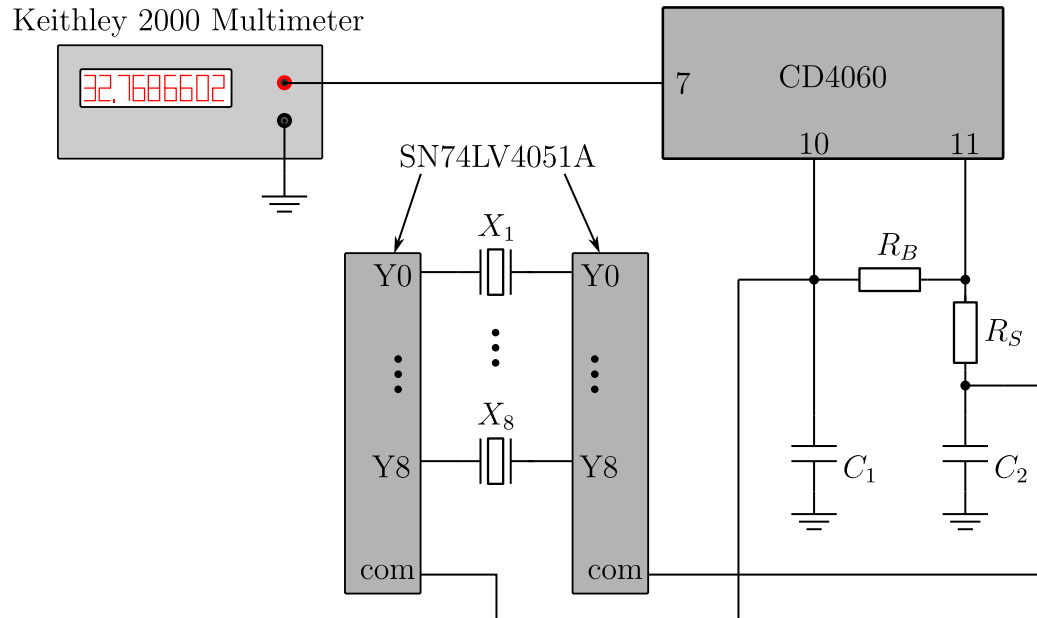


Figure 16: Multiplexer circuit used to drive eight crystals with one oscillator. Channel selection circuit and power supply are not shown.

4 Conducted Experiments

4.1 Parylene deposition on tuning forks

Hypothesis: The frequency of a quartz tuning fork decreases during deposition of parylene. The frequency shift is proportional to the thickness of the deposited film.

Two tuning forks, both soldered on a D-Sub connector (to facilitate their replacement), are coated with parylene (CVD001). Each tuning fork is connected to a separate Pierce oscillator circuit. Their initial frequencies differ by only 30mHz. Figure 17 shows the frequency shifts during deposition.

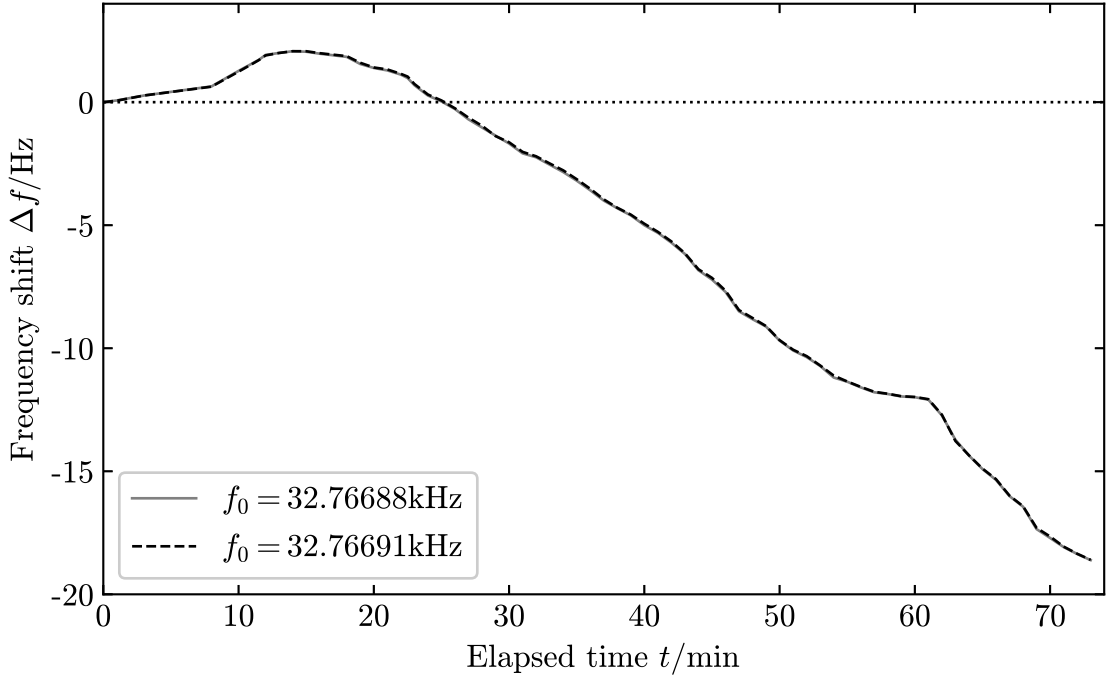


Figure 17: CVD001: The tuning forks, mounted together on the same base, have identical initial frequencies. They behave identically during deposition. At the beginning, the frequencies increase, and then drop linearly.

The frequency shifts of both tuning forks are identical. As soon as the butterfly valve is opened, the frequencies begin to change. At first, their frequencies increase slightly ($\Delta f \approx 2$ Hz), then decrease linearly after ~ 15 min. The frequencies stop changing immediately after the butterfly valve is closed.

4.2 Mutual interference of resonators

The fact that both tuning forks behave identically suggests mechanical coupling of the resonators. Huygens observed that two clocks mounted on a same wall would synchronize [11]. Since the tuning forks are rigidly mounted on the D-Sub connector, the oscillation of one tuning fork may propagate to the other, thus allowing the forks to synchronize.

Hypothesis: Mechanically isolated resonators behave differently during deposition due to specimen spread. Their initial frequencies differ slightly.

A new parylene deposition is undertaken (CVD002). This time, the oscillator circuits are placed inside the deposition chamber. The circuit boards each hang loosely from a wire fastened to the substrate holder, like clothes on a washing line. The tuning forks are soldered directly to their oscillator board. Their initial frequencies are now $\sim 2\text{Hz}$ apart, and shift differently (figure 18).

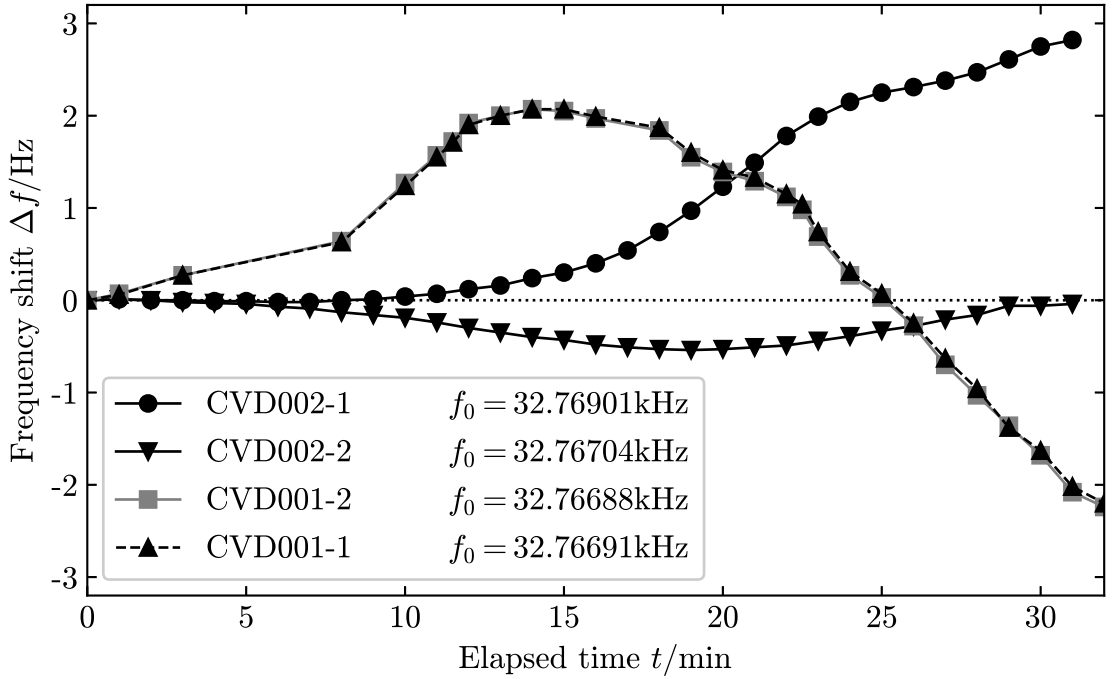


Figure 18: CVD002: Resonators rigidly mounted on the same base oscillate synchronously. Huygens observed a similar effect when two clocks are mounted on the same wall.

Mechanical coupling of the resonators most likely occurred during CVD001.

4.3 Investigation of the increase in frequency

During CVD002, none of the tuning forks' frequencies diminished. The deposited layer's thickness is estimated to be $d \approx 290\text{nm}$ (figure 19). During the first deposition, this film thickness was achieved at $t \approx 13\text{min}$. At that time, tuning fork frequencies were still rising. A certain amount of parylene may have to be deposited on the crystals before their frequencies drop.

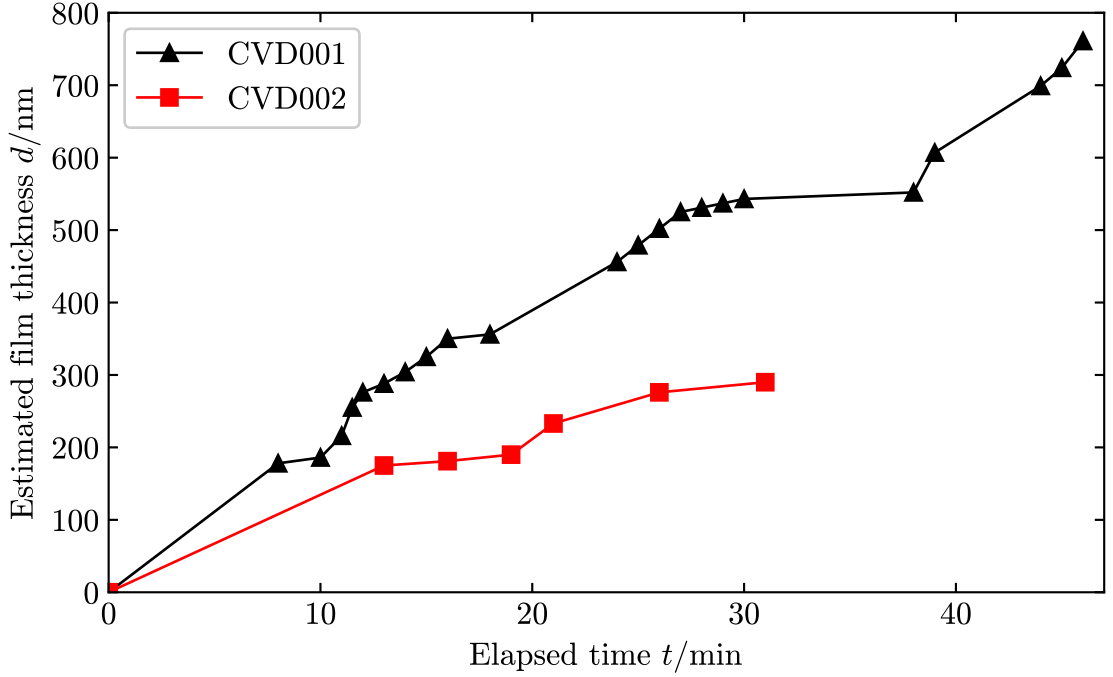


Figure 19: Deposition rate is significantly smaller during CVD002.

The increase in frequency although mass is added to the resonators is unexpected.

To investigate whether this may be due to inhomogeneous growth of the parylene layer on the resonators, ink is deposited with a permanent marker at different places on a tuning fork. After colouring one spot of the tuning fork, it is cleaned with propanol, and ink is deposited at another area. After cleaning the tuning fork, its frequency returns to its initial value. The experiment is repeated for several tuning forks.

A tuning fork's frequency increases by $\sim 2\text{Hz}$ when ink is deposited at its base between the prongs (figure 20). This is plausible, since this area is subjected to high flexure during oscillation (figure 9).

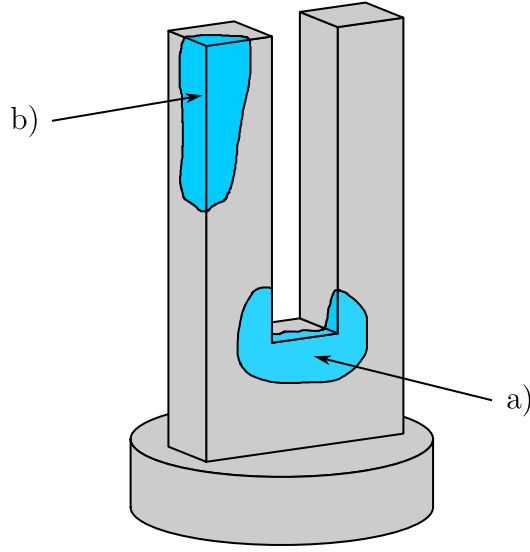


Figure 20: a) Tuning fork's frequency increases when ink is deposited at the base. b) Depositing ink on the tips causes the frequency to drop.

4.4 Parylene deposition on convex crystals

The tuning forks' frequency increase is believed to be due to their shape. Because of their different geometry and higher symmetry, the convex crystals may not have a “critical area”, at which added mass causes a positive frequency shift.

Hypothesis: The resonant frequency of convex crystals does not increase during deposition.

Two different 1.8432MHz convex crystals (ECS and IQD) are coated (CVD003). They are soldered to stiff wires, their oscillators are outside the deposition chamber. Tuning fork oscillator 2 is left in the chamber. The crystals are a few millimeters away from the substrate holder.

Similar behaviour as the tuning forks' is observed (figure 21).

When the frequencies of both crystals decrease, the slopes are similar. After ~ 68 min, the slopes increase significantly. The ECS crystal's slope increases slightly less than the IQD crystal's. According to figure 23, the sudden increase in slope is due to deposition rate increasing.

4.5 Resonator behaviour after first deposition

Hypothesis: Only “virgin” crystals behave anomalously at the beginning of deposition. After the parylene layer reaches a certain thickness, the resonance frequencies begin to

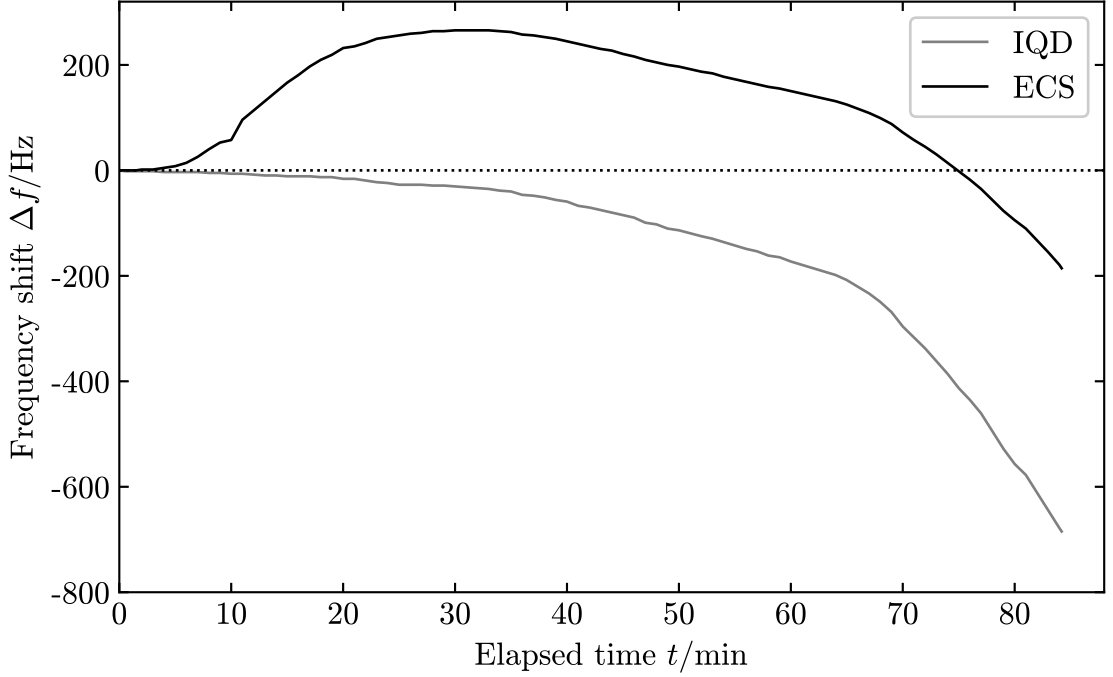


Figure 21: CVD003: Anomalous behaviour of the convex crystals is also observed.

decrease. This applies to the convex crystals, as well as to the tuning forks.

Two more depositions (CVD004 and CVD005) are conducted with the same setup. No reflector is placed in the deposition chamber during CVD005, as other samples are being coated, taking up all the space in the chamber. Reflected colours are observed on the samples.

Figure 22 shows the measured frequency shifts.

IQD crystal:

CVD004: The frequency shift slope is constant until $t \approx 50$ min, where it slightly decreases.

CVD005: The frequency shift is the same as during CVD004.

ECS crystal:

CVD004: Frequency shifts slower than the IQD crystal's frequency. At $t \approx 38$ min, frequency of the ECS crystal is ~ 80 Hz above the IQD crystal's. The slope increases slightly, and is similar to the IQD crystal's.

CVD005: The frequency shift is nearly identical to the IQD's during both depositions.

Tuning fork:

Frequency shifts during both depositions are identical. During CVD004, at $t \approx 50$ min, the slope decreases, like the IQD crystal's.

The slight decrease in slopes of the IQD crystal and tuning fork at $t \approx 50$ min must be due to deposition rate decreasing slightly. The slopes of the crystals during both depositions are all very similar.

The increase in frequency observed during the first depositions appears to occur only with “virgin” crystals. Once the crystals have recieved a sufficiently thick “primer” coat, their frequency shifts become proportional to the amount of deposited material.

4.6 Deposition rates

The linearity of the frequency shifts during CVD004 and CVD005 suggests a constant deposition rate during both depositions. However, the film growths shown in figure 23 are not linear. Deposition rate appears to decrease significantly once a thickness of $d \approx 280\text{nm}$ is reached.

A new series of depositions is undertaken (CVD006-CVD008). The substrate masking device is used to create parylene films of different thicknesses during each deposition.

The first version of the masking device is used during CVD006 and CVD007. No distinct areas are visible on the coated slides.

The improved version of the masking device is used for CVD008. The different areas are all clearly distinguishable from each other and reflect light in different colours. The thickness of each area is determined by measuring the reflectance spectrum with the spectrophotometer.

Hypothesis: Since the crystals' frequency shifts are linear during CVD, deposition rate must be constant.

Table 2 shows the layer thicknesses of the different areas, as well as the data used for the calculation. The film thicknesses are calculated using equations 42 and 43.

The mean thicknesses $\bar{d}(T)$ are calculated for all areas. The thickness of material deposited at the time t is calculated using:

$$d(t) = \bar{d}(T = 115\text{min}) - \bar{d}(T - t) \quad (44)$$

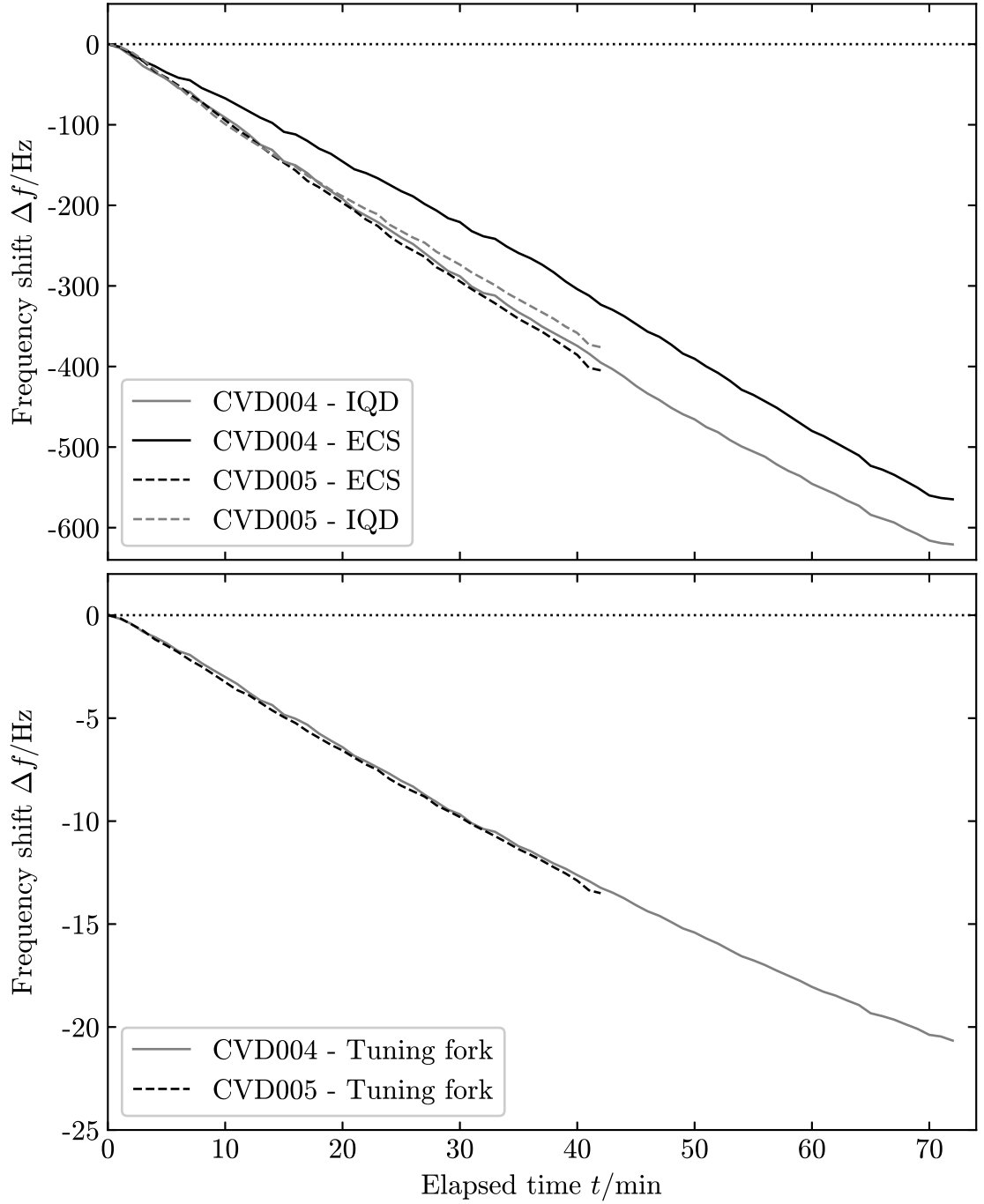


Figure 22: Once the parylene layer on the crystals has reached a certain thickness, their frequencies decrease linearly, indicating a steady deposition rate. The frequency shifts of two convex crystals and a tuning fork are shown during two deposition processes.

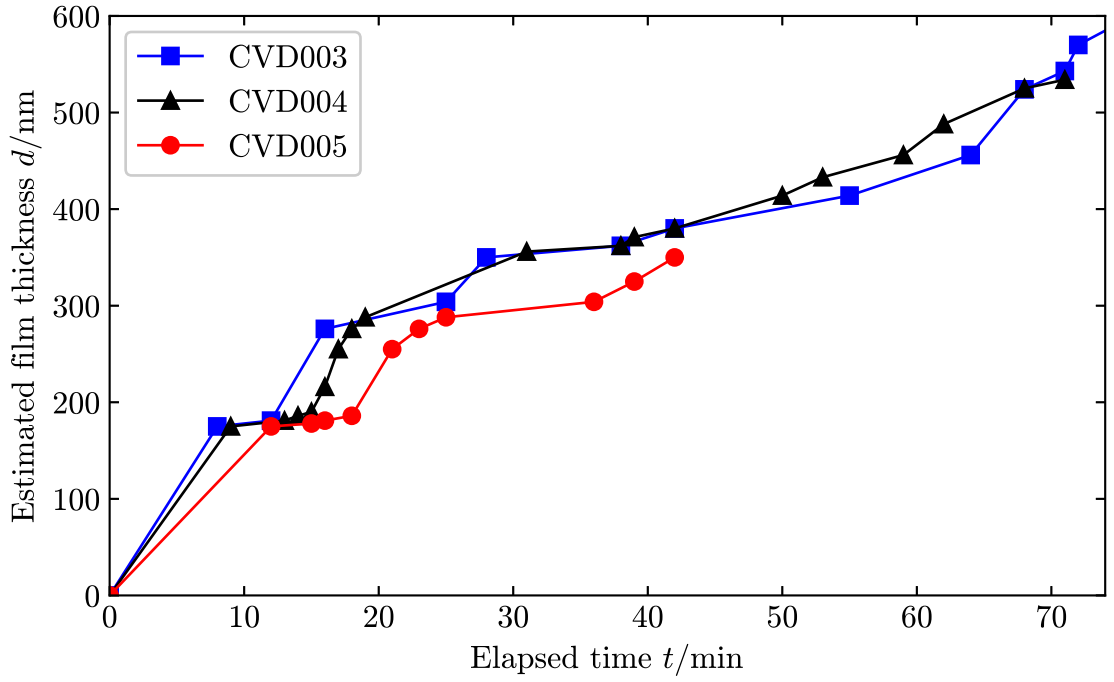


Figure 23: Deposition rate doesn't appear to be constant, although the quartzes' frequencies decrease evenly during CVD004 and CVD005. After $t \approx 68$ min deposition rate during CVD003 increases significantly, which explains the knee in figure 21.

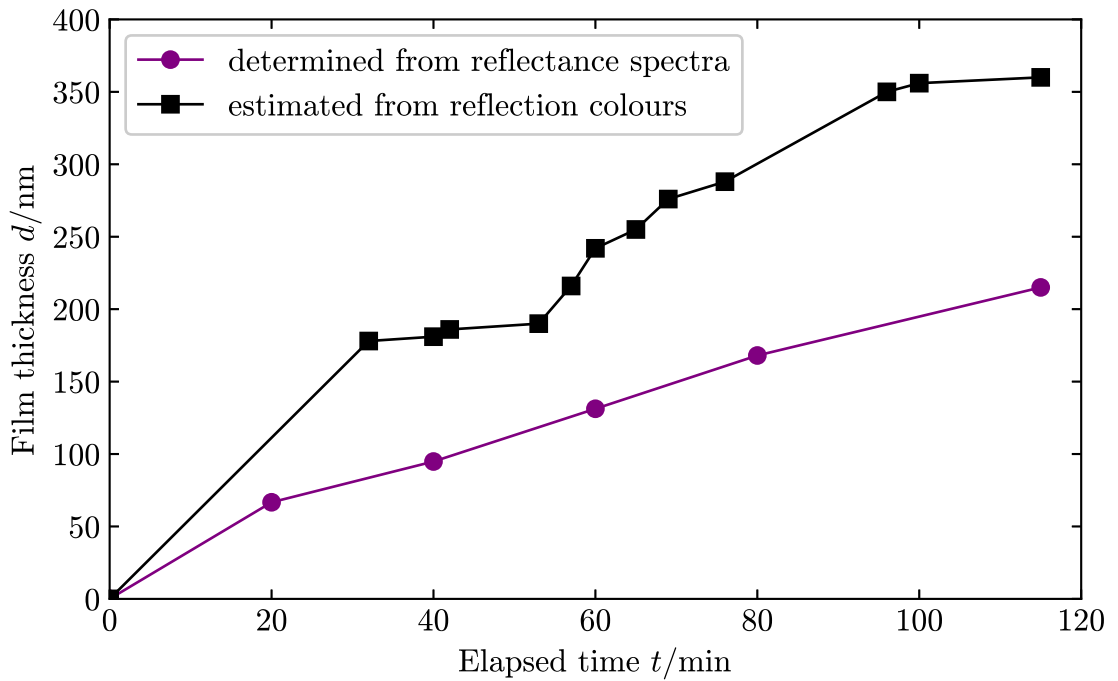


Figure 24: CVD008: The method used until now to monitor film thickness may be flawed. In reality, deposition rate is constant, and the film is much thinner than estimated.

Area	Exposed at t/min	Exposure time T/min	Peak/ Valley	Wavelength λ/nm	Interference order m	Thickness $d(T)/\text{nm}$
0	0	115	Valley	697	1	209.94
			Peak	467	2	210.994
			Valley	372	2	224.096
1	20	95	Peak	884	1	133.133
			Valley	514	1	154.819
			Peak	347.5	2	157.003
2	40	75	Peak	775	1	116.717
			Valley	410.5	1	123.645
3	60	55	Peak	501.5	1	75.5271
			Valley	305.5	1	92.0181
4	80	35	Peak	312	1	46.988

Table 2: CVD008: Film thicknesses of the different areas of the glass slide coated using the masking device.

$d(t)$ is plotted in figure 24 along with the film thicknesses estimated from the reflection colours during CVD. As in figure 23, the estimations suggest deposition rate varies during the process. However, the measured thicknesses show the film grows linearly during CVD. Furthermore, the final film thickness is significantly smaller than the estimate.

The thickness monitoring method used until now is believed to be flawed.

4.7 Specimen spread of convex crystals

During CVD005, the crystals' frequencies decreased with nearly identical slopes.

In order to compare the behaviour of several convex crystal specimens at once, the multiplexer circuit (figure 16) is used during CVD006-CVD008 to switch eight different crystals to the oscillator. Three new ECS and three new IQD crystals are connected, as well as the two crystals last coated during CVD005 (referred to as "reference" crystals). The crystals are orientated randomly at different heights in the chamber.

Hypothesis: All the crystals behave similarly after receiving their "primer" coats.

Evolutions of deposition rate during CVD006-CVD008 are more or less similar (figure 26).

CVD006 (figure 27): None of the new ECS crystals' frequencies increased during deposition, as was observed during CVD003. However, crystal behaviours are very different, the end frequencies spread over a range of ~ 800 Hz. The reference crystals' frequency shifts

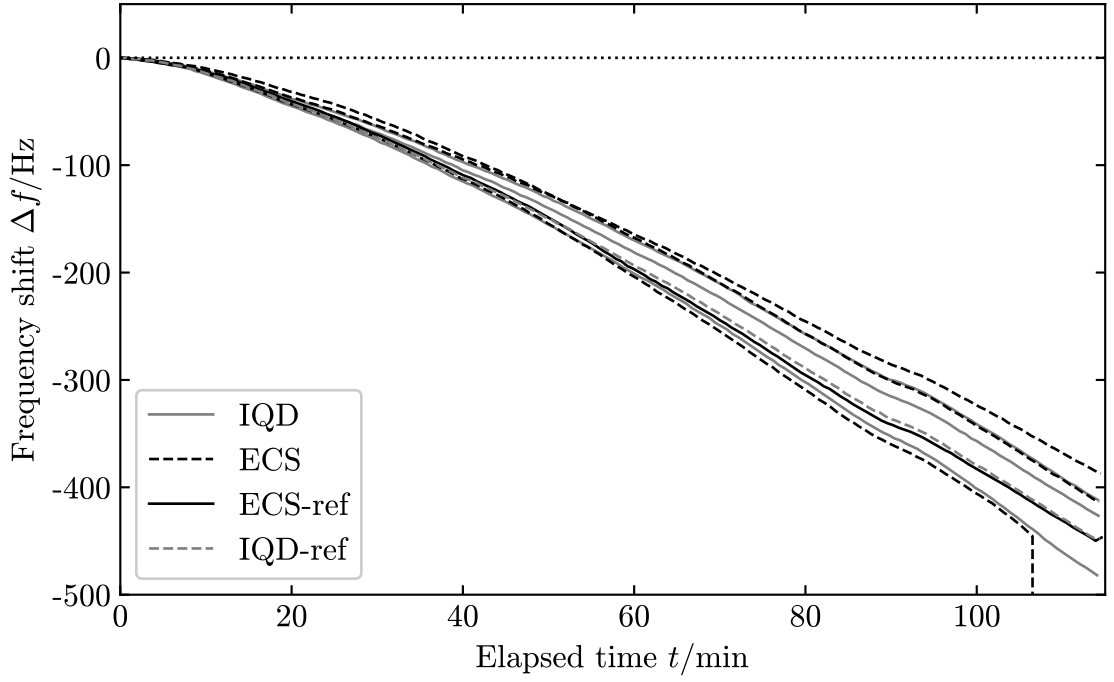


Figure 25: CVD008: Although the crystals are coated, their frequencies do not decrease linearly as soon as deposition begins.

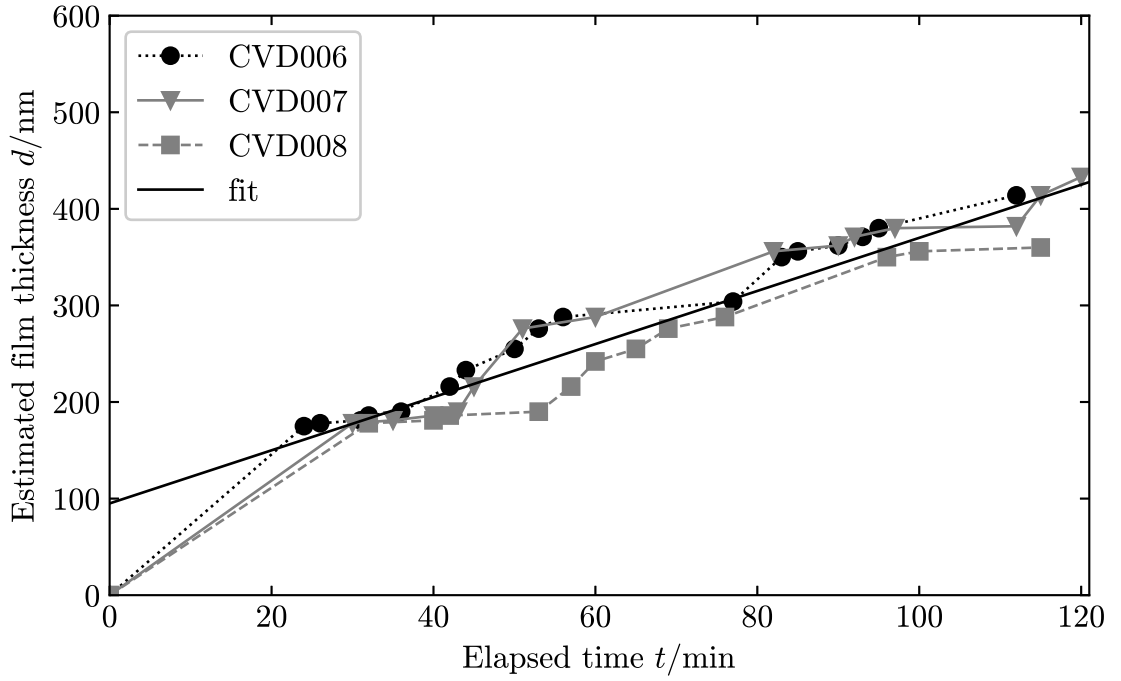


Figure 26: Deposition rates are similar during each of the three CVDs.

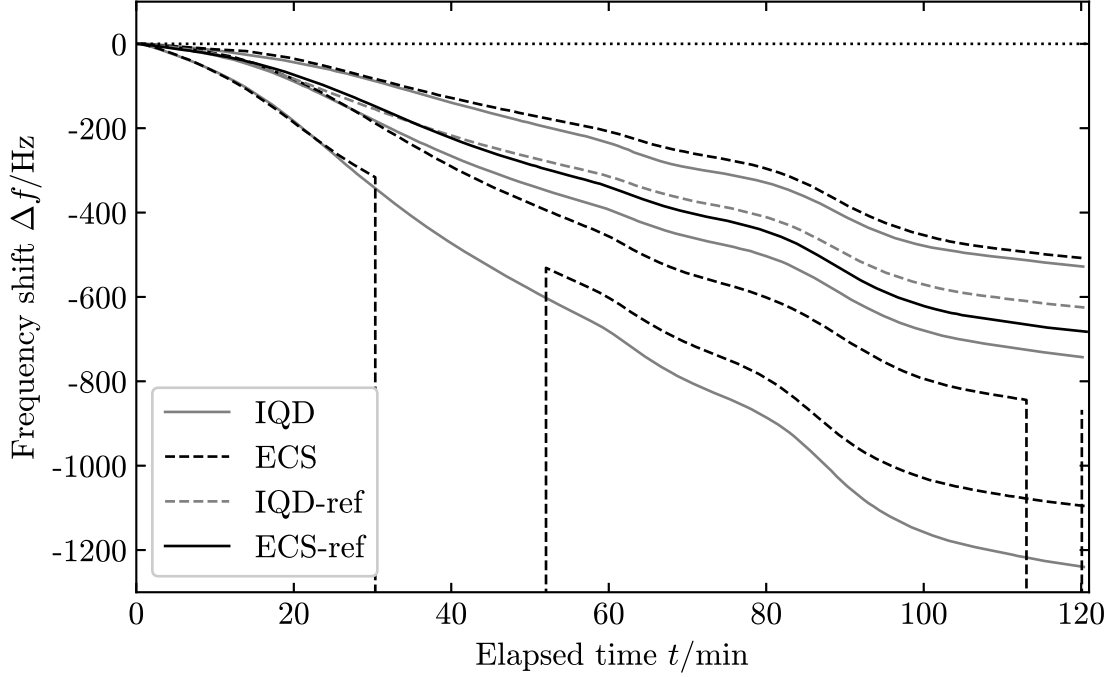


Figure 27: CVD006

are about ~ 30 Hz apart during the entire process. After $t \approx 100$ min, all the crystals' frequencies shift similarly (similar slopes). The “primer” coat must have reached a sufficient thickness at that time. Deposition rate appears to have fluctuated, since the reference crystals' curves are also wavy.

CVD007 (figure 28): All but two ECS crystals show similar frequency shifts. Their frequency shifts at deposition end spread over a range of ~ 80 Hz. Their slopes become constant at $t = 30$ min. The ECS crystals' frequency shifts lie ~ 100 Hz away from the other crystals'.

CVD008 (figure 25): Deposition rate was smallest during CVD008 (figure 26). The quartzes behave similarly as during CVD007. Their frequency shifts at deposition end spread over a range of ~ 100 Hz.

Figure 24 showed deposition rate to be constant, perhaps slightly higher for $t \lesssim 20$ min. However, frequencies shift slower at that time. Frequencies shift evenly only after $t \approx 40$ min.

This indicates that the substrate used to measure the thicknesses with the optical method hasn't been coated the same way as the crystals.

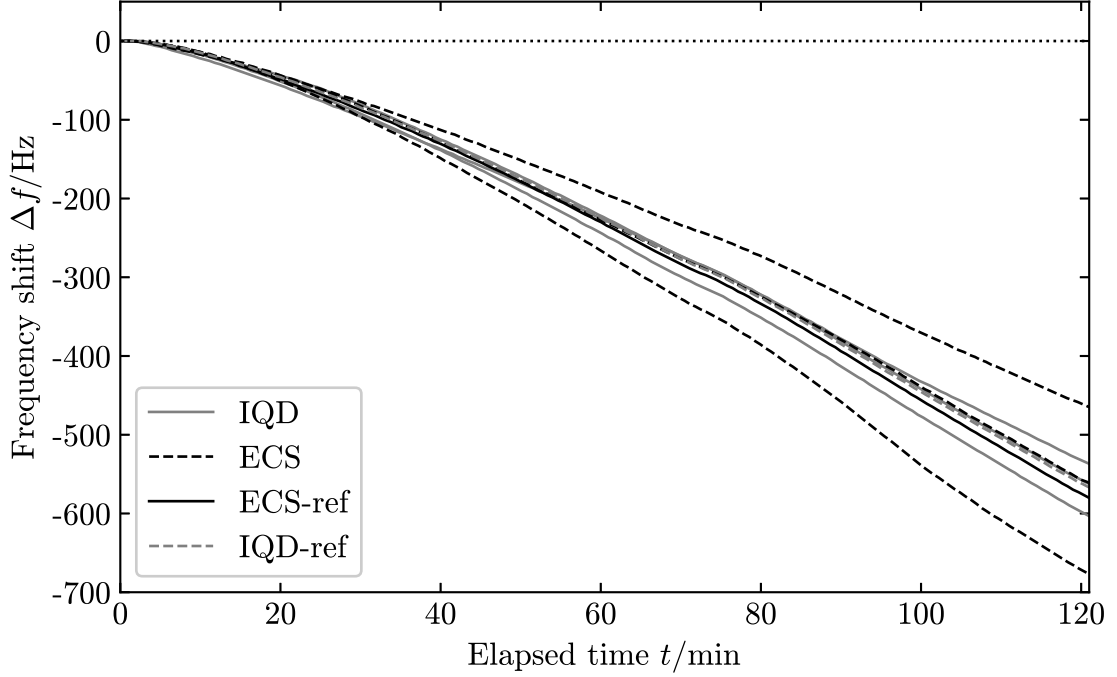


Figure 28: CVD007

Parylene deposition rate depends on the substrate's temperature. The substrate holder is cooled to a temperature of $T \approx 11^\circ\text{C}$ during deposition, while the crystals' temperature is about room temperature ($T \approx 23^\circ\text{C}$). This significant difference may very well result in a different deposition rate for the crystals.

5 Experimental Results

5.1 Optical thickness monitoring

The thicknesses of all deposited parylene films are determined by measuring the chromium reflectors' reflectance spectra. The thicknesses are obtained from reflectance maxima and minima wavelengths using equations 24 and 41. Table 3 shows the results, along with the reflector colour last observed and its corresponding wavelength.

The wavelengths λ of the reflectance minima and maxima are determined with an accuracy of $\Delta\lambda$, giving a thickness error of $\pm 5\%$:

$$\delta d = \frac{m}{2n} \cdot \Delta\lambda \quad (45)$$

$$\frac{\delta d}{d} = \frac{\Delta\lambda}{\lambda} \approx \frac{\pm 15 \text{ nm}}{300 \text{ nm}} = \pm 5\% \quad (46)$$

Use of the masking device during CVD008 allowed the film thicknesses at different times during deposition to be determined. This showed deposition rate to be more or less constant during a CVD process. The thickness at each colour change can be determined graphically using figure 24. These also figure in table 3.

CVD	Film thickness $d/\text{nm} \pm 5\%$	Colour	Wavelength λ/nm	Order m
008	84	yellow	580	1
	100	orange	605	1
	120	orange/red	620	1
	126	orange	650	1
	140	purple	415	2
	147	purple/blue	450	2
	160	blue	475	2
	189	green/yellow	570	2
	195	yellow	580	2
	215.0	yellow/orange	590	2
007	317.8	blue	475	3
006	293.2	purple/blue	450	3
005	?	green/yellow	570	2
004	376.3	yellow	580	3
003	614.0	red	650	4
002	133.7	blue	475	2
001	867.2	purple	415	6

Table 3: Parylene film thicknesses deposited during the conducted CVDs. The last observed reflected colour and its corresponding wavelength are given. m describes the number of times that colour was observed during deposition.

The measured thicknesses are significantly smaller than estimated (figures 23 and 26). The estimates were obtained using equation 24, assuming normal incidence ($\alpha = \beta = 0^\circ$), which is actually not the case. In practice, the incident angle α is about $\alpha \approx 30^\circ$.

In section 2.2.5, it has been shown that interference may occur when one of three different conditions are fulfilled (equations 21 to 23). From these, three different formulas for film

thickness can be derived: Equations 24 to 26 are plotted in figure 29. The angle of incidence is not neglected.

The data gathered in table 3 is layed over figure 29.

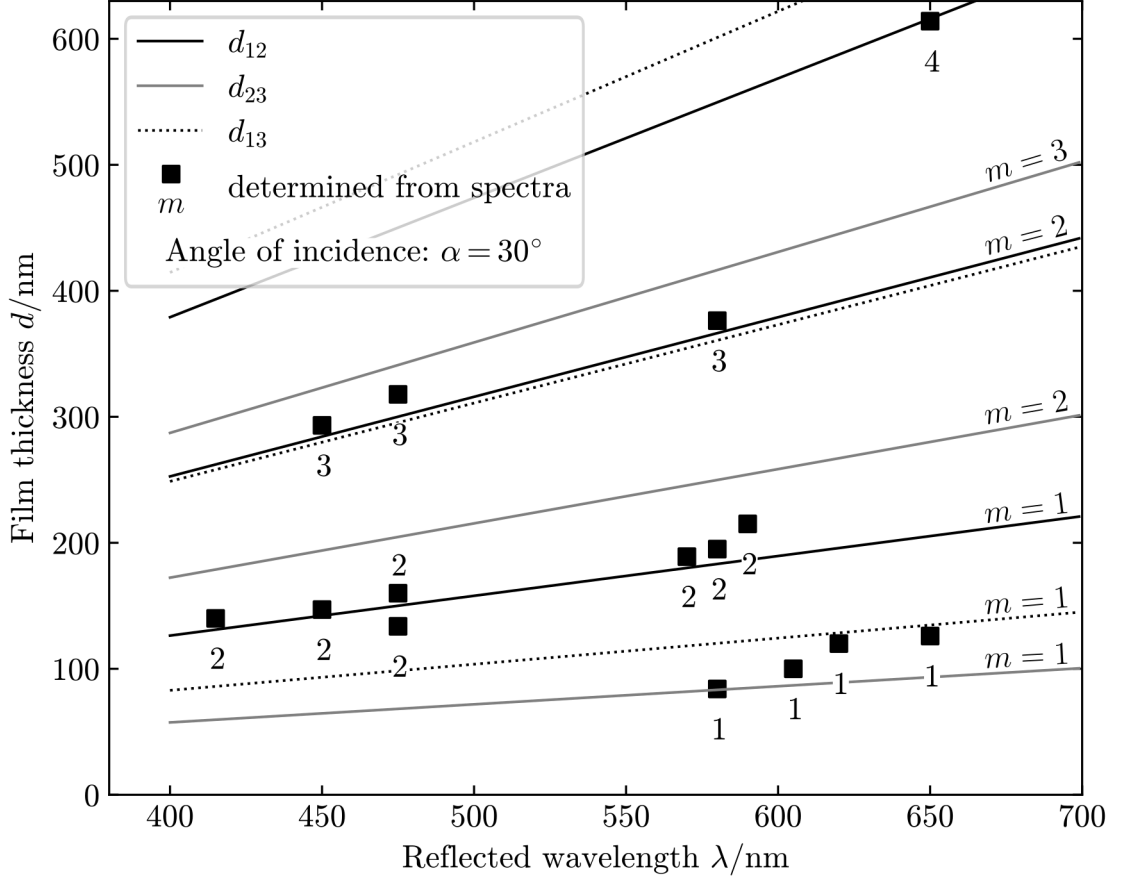


Figure 29: Equation 24 gives correct results for the film thickness. However, the reflected colours observed first do not correspond to the first interference order.

The film thickness can indeed be determined from the reflection colours observed in the sample using equation 24. However, equation 24 is only applicable beginning at the second colour sequence. The reflected colours observed first arise from interference between beams reflected multiple times inside the parylene film ($\vec{k}_2$ and $\vec{k}_3$ in figure 4). Here, equations 26 or 25 apply. This makes perfect sense, since the film is not thick enough for beam $\vec{k}_2$ to accumulate enough phase shift to interfere with $\vec{k}_1$.

The optical film thickness monitoring method is absolutely viable. Different colours lie roughly $\lesssim 30$ nm apart. Using that as the wavelength uncertainty of the human eye gives a thickness error (equation 46) of 10% at worst. This method is quite accurate, provided

the incident angle of the light source is taken into account.

5.2 Quartz crystal microbalance

Expected frequency shift per nanometer parylene deposited on the crystal F_q , calculated using the Sauerbrey equation for the 1.8 MHz convex crystals:

$$\left(\frac{\Delta f}{\Delta d}\right)_{crystal} = \frac{f_0^2}{N} \cdot \frac{\rho_p}{\rho_q} = \frac{(1.8432 \cdot 10^6 \text{ Hz})^2}{1.670 \cdot 10^3 \text{ Hz} \cdot \text{m}} \cdot \frac{1.11}{2.65} \cdot \frac{10^{-9} \text{ m}}{1 \text{ nm}} = 0.852 \frac{\text{Hz}}{\text{nm}} \equiv \hat{F}_q \quad (47)$$

The frequency shift per nanometer parylene deposited on the sample F is expected to be twice that, because the quartz is coated on both sides:

$$\frac{\Delta f}{\Delta d} = 1.70 \frac{\text{Hz}}{\text{nm}} \equiv \hat{F} \quad (48)$$

This constant doesn't change much when the initial frequency f_0 changes:

$$\frac{\delta F}{F} = 2 \frac{\Delta f_0}{f_0} < 10^{-3} \quad (49)$$

Therefore, F need not be calculated for the exact initial frequency of the crystal.

During CVD of parylene, the thickness of parylene deposited onto the quartz does not necessarily correspond to the layer thickness on the substrate. This is due to temperature differences between crystal and substrate, which have an effect on deposition rate. F determined from experimental data will therefore contain a factor C which depends on the physical setup of the quartz crystal microbalance:

$$F = C \cdot \hat{F} \quad C > 1 \quad (50)$$

Table 4 shows the data measured during the depositions, from which F is calculated.

Errors:

$$\Delta \left(\frac{\Delta f}{\Delta d}\right) = \frac{\delta d}{d} \cdot \left(\frac{\Delta f}{\Delta d}\right) = \pm 5\% \cdot \left(\frac{\Delta f}{\Delta d}\right) \quad (51)$$

The calculated F are greater than $\hat{F}$, as expected, and differ from one another between depositions.

CVD	Film thickness $d/\text{nm} \pm 5\%$	Frequency shift $\Delta f/\text{Hz}$	$F =$ $ \frac{\Delta f}{\Delta d} \pm 5\% = \left[\frac{\text{Hz}}{\text{nm}} \right]$	Crystal
004	376.3 ± 18.9	564.8 620.80 20.7	1.50 ± 0.08 1.65 ± 0.09 $(55.0 \pm 2.8) \cdot 10^{-3}$	ECS IQD Tuning fork
005	$180.04 \pm 9.1^*$	404.8 376.0 13.5	2.25 ± 0.04 2.09 ± 0.04 $(75.0 \pm 1.4) \cdot 10^{-3}$	ECS IQD Tuning fork
007	317.8 ± 15.9	583.4 569.26 677.6 564.29 466.5 604.3 563.7 538.5	1.84 ± 0.10 1.79 ± 0.09 2.13 ± 0.11 1.78 ± 0.09 1.47 ± 0.08 1.90 ± 0.10 1.77 ± 0.09 1.69 ± 0.90	ECS-ref IQD-ref ECS ECS ECS IQD IQD IQD
008	215.0 ± 10.8	446.8 447.4 ? 413.5 387.10 481.8 411.7 426.5	2.08 ± 0.11 2.08 ± 0.11 ? 1.92 ± 0.10 1.80 ± 0.10 2.24 ± 0.12 1.92 ± 0.10 1.98 ± 0.10	ECS-ref IQD-ref ECS ECS ECS IQD IQD IQD

Table 4: Frequency shift per nanometer parylene deposited on the sample varies between runs. *) CVD005 film thickness is calculated from the last observed reflected wavelength.

During CVD007, F varies the most between crystals, with $\Delta F = 0.4 \text{Hz} \cdot \text{nm}^{-1}$, and the average being $\overline{F}_7 = 1.8$.

The average of all F is

$$\overline{F} = (1.9 \pm 0.5) \text{Hz} \cdot \text{nm}^{-1} \quad (52)$$

And the factor C is

$$C = \frac{\overline{F}}{\hat{F}} = \frac{1.9 \pm 0.5}{1.7} = 1.1 \pm 0.3 \quad (53)$$

Having measured a certain frequency shift Δf during a parylene deposition, the thickness

of the film can be determined (perhaps “grossly estimated” is more fitting) to an accuracy of

$$\delta d = \pm d \cdot \frac{\Delta \bar{F}}{\bar{F}} = \pm d \cdot 0.27 \quad (54)$$

For a 350 nm thick parylene film, as is required at the THz group, this uncertainty corresponds to $\delta d = 95$ nm.

The fact F differs between depositions is partly due to the position of the crystals being changed. For example, CVD005 was conducted to deposit an insulator layer on a batch of samples. The crystals had to be moved to make room for the samples.

The setup stayed unchanged during CVD007 and CVD008. However, F still increased in CVD008. This may imply that F evolves when the crystal is coated. Further depositions should be undertaken to examine this issue and to determine the factor C . Mounting the crystals at a fixed spot close to the substrate holder would be good start.

Conducting a parylene deposition without cooling the sample would minimize the temperature difference between crystals and sample. The factor C would be expected to become $C \approx 1$, and F could be compared with the theoretical value $\hat{F} = 1.70 \text{ Hz} \cdot \text{nm}^{-1}$.

6 Conclusions

In the course of this bachelor thesis, the thickness of parylene films deposited via chemical vapour deposition was determined using three different methods.

Using a spectrophotometer, the reflectance spectra of parylene coated microscope slides were measured. Film thickness was determined with an accuracy of 5% from the wavelengths of reflection maxima and minima.

The other two methods allow in-situ monitoring of the film thickness during deposition.

By observing the colour of light reflected in the samples during deposition, wavelengths of reflectance maxima can be determined by eye. An accuracy of 5 – 10% is achievable, depending on the colour. This method was used until now at the THz group. It was found to yield reasonable results only when the first sequence of reflected colours is neglected. To obtain accurate results, the incident angle of the light source directed at the samples should be taken into account.

A simplistic quartz crystal microbalance can be set up using quartz crystals obtained from electronics suppliers.

1.8432 MHz crystals and 32.768 kHz tuning forks were coated with parylene. During deposition, their frequencies decreased in proportion to the amount of parylene deposited. This was only the case when the crystals' parylene coat reached a thickness of $d_0 \approx 300$ nm. The cause of this could not be determined. The proportionality constant $F = \frac{\Delta f}{\Delta d}$ was found to vary between depositions. This is believed to be due to the crystals' positions inside the chamber being changed. However, it cannot be ruled out that F evolves with the parylene layer thickness coating the crystals. While the sample was being cooled during deposition, the crystals stayed at room temperature, causing less parylene to be deposited on the crystals. F was determined from the film thickness on the sample. The obtained F differed from that determined by the Sauerbrey equation $\hat{F}$:

$$\overline{F} = (1.9 \pm 0.5) \text{ Hz} \cdot \text{nm}^{-1} > \hat{F} = 1.7 \text{ Hz} \cdot \text{nm}^{-1}$$

This result does not allow precise film thickness determination. Further experiments must be conducted. Carrying out a parylene deposition without cooling the sample and mounting the crystals near the substrate holder is suggested to obtain equal layer thicknesses on the sample and the crystals.

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Eigenständigkeitserklärung

I, Matthieu Bernard Michel Mézière, hereby confirm that this thesis was written independently by myself without the use of any sources beyond those cited, and all passages and ideas taken from other sources are cited accordingly.

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